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"Vegetation and Energy"
"CGLOPS-1"
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ALGORITHM THEORETICAL BASIS DOCUMENT

EVAPOTRANSPIRATION AND HEAT FLUX 300M

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List of Acronyms

AC	Atmospheric Corrections
AD	Applicable document
ATBD	Algorithm Technical Basis Document
CAMS	Copernicus Atmosphere Monitoring Service
CDSE	Copernicus Data Space Ecosystem
CEOS	Committee on Earth Observation Satellites
CGLOPS	Copernicus Global Land Operations
CLMS	Copernicus Land Monitoring Service
COG	Cloud Optimized Geotiff
CRO	Cropland plant functional type
CSH	Closed Shrubland plant functional type
CV	Coefficient of Variation
DEM	Digital Elevation Model
DMS	Data Mining Sharpener
EBC	Energy Balance Closure
EC	Eddy Covariance
EO	Earth Observation
ETA	Evapotranspiration - Actual
EUMETSAT	European Organisation for the Exploitation of Meteorological Satellites
FAO	Food and Agriculture Organisation of the United Nations
FWC	Framework Contract
G	Ground heat flux
GBOV	Ground-Based Observations for Validation
GCOS	Global Climate Observing System
GEDI	Global Ecosystem Dynamics Investigation
GLEAM	Global Land Evaporation Amsterdam Mo
GRA	Grassland plant functional type
H	Sensible heat flux
HF	Heat Flux
ICOS	Integrated Carbon Observation System
JRC	Joint Research Centre
LAI	Leaf Area Index
LE, λE	Latent heat flux

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LIDF	Leaf Inclination Distribution Function
LOPEX	Leaf Optical Properties Experiment
LPV	Land Production Validation
LST	Land Surface Temperature
MNDWI	Modified Normalized Difference Water Index
MODIS	Moderate Resolution Imaging Spectroradiometer
NetCDF	Network Common Data Form
NIR	Near InfraRed
NMDI	Normalized Difference Moisture Index
NRT	Near Real Time
OLCI	Ocean and Land Colour Instrument
OSH	Open Shrubland plant functional type
PAR	Photosynthetically Active Radiation
PDU	Product Data Unit
PSRI	Plant Senescence Reflectance Index
PFT	Plant Functional Type
PUM	Product User Manual
Rn	Net radiation
RMSE	Root Mean Squared Error
RTM	Radiative Transfer Model
SAA	Sun Azimuth Angle
SAV	Savanna plant functional type
S2GM	Sentinel-2 Global Mosaic
SEN-ET	Sentinels for Evapotranspiration
SLSTR	Sea and Land Surface Temperature Radiometer
SSEBop	operational Simplified Surface Energy Balance
SWI	Soil Water Index
SZA	Solar Zenith Angle
TOC	Top Of Canopy
TSEB	Two-Source Energy Balance
VAA	View Azimuth Angle
VR	Validation Report
VZA	View Zenith Angle
WaPOR	WAter Productivity through Open-access of Remotely sensed derived data
WET	Wetland plant functional type

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WSA	Wooded Savanna plant functional type

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EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified global bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, water cycle, energy budget and terrestrial cryosphere. Production and delivery take place in a timely manner and are complemented by the constitution of long-term time series.

This Algorithm Theoretical Basis Document (ATBD) describes the two modelling frameworks used to derive land water and energy fluxes (i.e. actual evapotranspiraton (ETA) and Heat flux), including all the input variables and parameters needed. The retrieval methods are based on two scientifically sound approaches: Sen-ET using the TSEB model and WaPOR using the ETLook model.

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1 BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVES

This document describes the physical and mathematical basis of the algorithm applied to generate the ETA and Heat Flux (HF) v1.0 product in the Copernicus Land Monitoring Service. The ETA product consists in dekadal gap-filled evapotranspiration, plant transpiration and soil evaporation. The HF v1.0 product consists in clear-sky instantaneous sensible and latent heat fluxes.

This ATBD aims to serve as a reference document for implementing the production chain.

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows.

- Chapter 1 introduces the document.
- Chapter 2 recalls the requirements, including the expected performances.
- Chapter 3 describes the retrieval methodology.
- Chapter 4 lists the references cited in this document

1.3 RELATED DOCUMENTS

1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

Available at https://etendering.ted.europa.eu/cft/cft-display.html?cftId=13795

1.3.2 Input

Document ID	Descriptor
CGLOPS1_PUM_S3-TOC-Reflectance300m-V2.3	Product User Manual of the Sentinel-3 Top-Of-Canopy reflectances V2.3 products

Available at: https://land.copernicus.eu/en/technical-library

1.3.3 Output

Document ID	Descriptor
CGLOPS1_PUM_ETA300-V1	Product User Manual summarizing all information about the ETA300 V1.0 product.
CGLOPS1_VR_ETA300-V1	Quality Assessment and Validation Report of ETA300 V1.0 product

Documents available at https://land.copernicus.eu/en/technical-library

1.3.4 External document

Document ID	Descriptor
OMPC.ACR.HBK.002	Copernicus Sentinel-3 SLSTR Land User Handbook

https://sentiwiki.copernicus.eu/__attachments/1672112/OMPC.ACR.HBK.002 - Sentinel 3 SLSTR Land Handbook 2024 - 1.4.pdf

ATBD_S2ToolBox

https://step.esa.int/docs/extra/ATBD_S2ToolBox_V2.0.pdf

WaPOR Data Manual - Evapotranspiration v2

https://bitbucket.org/cioapps/wapor-et-look/downloads/FRAME_ET_v2_data_manual_finaldraft_v2.2.pdf

COP-DEM_GLO-90-DTED	Copernicus Global Digital Elevation Model. European Space Agency, 2024.

Distributed by OpenTopography. https://doi.org/10.5069/G9028PQB

2 REQUIREMENTS

According to the applicable document [AD1], the requirements relevant for the ETA product are described below.

2.1 SPECIFIC TECHNICAL DETAILS AND REQUIREMENTS

PRODUCT SPECIFICATION	
Geometric Properties:	
Baseline dataset pixel resolution	300m
Target baseline location accuracy	Better than 0.5 pixels
Coordinate position	Centre of the pixel
Geodetical datum	WGS84
Geographic projection	Regular latitude/longitude grid
Geographic coverage:	Global
Temporal resolution	10-day period (dekad: days 1-10, 11-20, 21 end of month)
Timeliness	Within 2 days (optimally 1 day) after the end of each dekad

ETA - EVAPOTRANSPIRATION	
Evapotranspiration is a key component of hydrological balance. Evapotranspiration is the sum of evaporation and plant transpiration from the Earth's land and ocean surface to the atmosphere. Foreseen products: sensible heat flux, latent heat flux, bare soil evaporation, interception loss, transpiration.	
Uncertainty (2-sigma)	https://gcos.wmo.int/en/essential-climate-variables/evaporation
Stability (per decade i.e. 10 years)	Threshold: 10% Goal: 5%

2.2 FURTHER REQUIREMENTS

2.2.1 Output product composition

Products may contain various information layers and ancillary information – the base reference for product packages are the operational products as on 01.03.2023.

2.2.2 Data Structure

Data coding[1] shall be compatible with the Global Land products as on 01.03.2023 and/or follow the INSPIRE specifications, where applicable.
Ancillary information shall be as currently used and include at least the following:

- The number of measurements per pixel used to generate any synthesis product
- The per-pixel date of the individual measurement or the start-end dates of the period covered
- Quality indicators, with explicit per-pixel identification of the cause of anomalous parameter result.

The product naming and filename conventions that are used in the Copernicus Global Land component production as on 01.03.2023 shall be followed. This may be adapted for complete product collections upon agreement with the contracting authority during the Framework.

2.2.3 Data format

To ensure interoperability with the current Global Land component (operational product data formats and archive data formats) and other Copernicus services, all datasets will be available in NETCDF. Additional format such as Cloud Optimized Geotiff (COG) or ZAR format can be proposed for production or could be requested by the Contracting Authority during the Framework Contract.

2.2.4 Uncertainties and Validation

Uncertainties indicated in the product specifications above follow the threshold and goal proposed by GCOS#245.

Uncertainties estimates should account for the error propagation uncertainty coming from input data though the retrieval algorithms as during the contract period, ESA plans to imbed uncertainties in the Sentinel-3 ground segment products and, for Sentinel-2, there is an offline tool to determine Level 1 uncertainties; these can be used for propagation in the production chain.

[1] Data coding is the provision for the number of bits, range of values, usage of reserved values, content of status map, conversion to physical values formulas.

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Validation of the products shall conform to at least the CEOS LPV standards. Wherever appropriate the bio-geophysical variables shall be validated and compared to CEOS CAL/VAL data sets and/or Ground-Based Observations for Validation (GBOV) of Copernicus Global Land component of Copernicus Global Land component when biophysical parameters are available.

2.2.5 Input data

Copernicus sentinel data are available from https://dataspace.copernicus.eu

Bio-geophysical variable products should be based on common base reflectance data:

- Sentinel 3: Derived Top-of-Canopy reflectance may be brokered or produced as it is the case on 01.03.2023 under the CGLOPS contracts;
- Sentinel 2: the Global Land Sentinel-2 Global Mosaic (S2GM)[2] component provides temporal mosaic of surface spectral bands that can be brokered and/or directly used.

Up to and including 2019, products archives of the Global Land Component have been based on SPOT VGT, Proba-V, ENVISAT, MODIS, TOPEX/Poseidon, Jason-1, Jason-2, Jason-3, datasets, which are available through the https://dataspace.copernicus.eu.

LST and SWI can be based on geostationary and other satellite data.

Ancillary satellite data that is purchased through Copernicus and put at disposal of the Services is available through the Data Warehouse and will become available on the Copernicus Dataspace Ecosystem.

Ancillary data sets, other than satellite imagery described above, that might be required shall be the responsibility of the contractor.

2.2.6 Product delivery

Products shall be delivered to the Copernicus Land Component dissemination. The Copernicus Data Space Ecosystem infrastructure will be used to provide access to the final map products.

[2] https://land.copernicus.eu/en/products/global-image-mosaic

3 METHODOLOGY DESCRIPTION

3.1 OVERVIEW

The estimation of actual evapotranspiration (ETA), or latent heat flux (λE) in terms of energy, can be obtained from the energy balance equation (Eq. 1):

$$\lambda E \approx R_n - G - H \quad \text{Eq. 1}$$

where G is the heat transmitted and stored into the ground, R_n is the net radiation, and H is the sensible heat flux. Therefore, in order to retrieve ETa (or λE[3]) we must derive the other components of the energy balance (R_n, G, and H), which mainly depend on surface temperature, plant biophysical traits, weather, and canopy structure.

3.1.1 TSEB-PT model

The Two-Source Energy Balance (TSEB) modelling scheme was proposed by Norman et al (1995) and afterwards refined and applied in a multitude of applications and studies (Anderson et al., 2024) including in the Sen-ET framework (Guzinski et al., 2020) and implemented as open source in pyTSEB Python package (Nieto, 2023). In this modelling scheme the directional radiometric Land Surface Temperature (LST) ($T_R(\theta)$) is split into the temperatures of vegetation and soil, based on the vegetation cover, and LST observation geometry (Eq. 2).

$$T_R(\theta) \approx [f(\theta)T_C^4 + [1 - f(\theta)]T_S^4]^{0.25}$$
$$f(\theta) = 1 - exp\left(\frac{-0.5\Omega(\theta)LAI}{cos\theta}\right) \quad \text{Eq. 2}$$

[3] Both λE and LE are used indistinctly in literature, indeed λE is usually typed in energy balance formulations in order to reflect that water evaporation (E) is converted to heat fluxes based on the latent heat of water vaporization (λE), whereas LE is usually typed within statements and paragraphs

where θ is the view zenith angle of the thermal observation and Ω(θ) is the clumping factor of the vegetation at view angle θ (Kustas and Norman, 1999) and has a value of less than 1 for clumped vegetation and subscripts C and S denote canopy and soil respectively.

Based on this split, the energy fluxes of vegetation and soil (net radiation - R_n, sensible heat flux - H, latent heat flux - λE, soil heat flux - G) are estimated separately, before being combined to obtain the bulk surface fluxes (Eq. 3).

$$\begin{aligned} R_{n,C} &= H_C + \lambda E_C \\ R_{n,S} &= H_S + \lambda E_S + G \\ R_n &= (H_C + H_S) + (\lambda E_C + \lambda E_S) + G = H + \lambda E + G \end{aligned} \qquad \text{Eq. 3}$$

The soil (canopy) sensible heat flux is computed from the gradient between the soil (canopy) temperature (T_S and T_C respectively) and the air temperature at the sink-source height. This transfer of heat between the two components and the atmosphere is modulated by resistances to heat exchange organized in a series resistance network (in analogy to electrical systems) which depend on aerodynamic and meteorological conditions.

Since there are multiple solutions to T_C and T_S satisfying Eq. 2, an iterative approach is employed. The initial assumption is that green canopy transpires at potential rate based on Priestley-Taylor formulation (Eq. 4) (Priestley and Taylor, 1972):

$$\lambda E_C = \alpha_{PT} f_g \frac{\Delta}{\Delta + \gamma} R_{n,C} \qquad \text{Eq. 4}$$

where α_{PT} is the Priestley-Taylor coefficient, Δ is the slope of the vapour pressure to air temperature curve (mbar.K^{-1}) and γ is the psychrometric constant (mbar.K^{-1})).

For short-canopies (herbaceous vegetation, shrublands and savannas) α_{PT} has an initial value of 1.26, while for tress, α_{PT} is lowered to according to Komatsu (2005) in order to account for reduction in stem conductivity (and therefore stomatal conductance). In the case of broadleaved forest, a constant value of 0.82 is given while for conifer forests, α_{PT} decreases with canopy height (*hc*) like in Eq. 5 (Komatsu 2005).

$$\alpha_{PT} = -0.269ln(hc) + 1.31 \quad \text{Eq. 5}$$

If unrealistic fluxes are obtained ($\lambda E_C < 0$ and $\lambda E_C < 0$) then the canopy transpiration (i.e., α_{PT}) is sequentially reduced and soil and canopy temperatures and fluxes are recalculated until realistic values are obtained. This implementation of the TSEB scheme is called the TSEB-PT model.

TSEB-PT outputs instantaneous fluxes at the time of thermal image acquisition. The modelled instantaneous latent heat flux (λE_{inst}), calculated during clear-sky conditions, is extrapolated to daily ET values as $\lambda E_{daily} = \lambda E_{inst} \frac{S^{\downarrow}_{daily}}{S^{\downarrow}_{inst}}$ with $S^{\downarrow}_{daily}$ and $S^{\downarrow}_{inst}$ are the daily and instantaneous shortwave irradiances, respectively.

3.1.2 ETLook model

ETLook model (Bastiaanssen et al., 2012) is used in the WaPOR framework and is described in detail in Section 5 of "WaPOR Data Manual - Evapotranspiration v2". Similarly to TSEB-PT, ETLook is a two-source model, meaning that it derives soil evaporation (E) and vegetation transpiration (T) as two separate fluxes, and it ensures conservation of energy at the land-surface. The model assumes potential rates of daily E and T based on the Penman–Monteith equation (Monteith, 1965) that are throttled down to actual E and T using stress factors (Eq. 6):

$$E = \frac{\Delta\left(R_{n,S} - G\right) + \rho c_p \frac{\Delta_e}{r_{a,S}}}{\Delta + \gamma\left(1 + \frac{r_S}{r_{a,S}}\right)}$$
$$T = \frac{\Delta\left(R_{n,C}\right) + \rho c_p \frac{\Delta_e}{r_{a,C}}}{\Delta + \gamma\left(1 + \frac{r_C}{r_{a,C}}\right)} \quad \text{Eq. 6}$$

where Δe (mbar) is vapor pressure deficit, ρ (kg.m^{-3}) is the air density, c_p (J.kg^{-1}.K^{-1}) is specific heat of dry air, $r_{a,S}$ and $r_{a,C}$ are aerodynamic resistances for soil and canopy respectively and r_S and r_C are resistances of soil and canopy. All resistances are in s.m^{-1}.

The resistances of soil and canopy to energy transfer are calculated taking the stress factors into account (Eq. 7):

$$r_S = b\left(S_e^{top}\right)^c$$
$$r_C = \frac{r_{s,min}}{LAI_{eff}} \frac{1}{S_t S_v S_r S_m} \quad \text{Eq. 7}$$

where *b* and *c* are soil resistance parameters, $r_{s,min}$ (s.m^{-1}) is the minimum stomatal resistance, LAI_{eff} is effective leaf area index. Soil evaporation is limited by top-soil moisture (S_e^{top}), while plant transpiration is affected by air temperature stress (S_t), vapour pressure stress (S_v), radiation stress (S_r) and root-zone soil moisture stress (S_m) (Jarvis, 1976).

The soil moisture required for E and T stress factors is derived using a trapezoid constructed in the LST - vegetation fractional cover (f_C) space (Yang et al., 2015). The trapezoid corner values as set based on theoretical calculations by inverting the Penman-Monteith equation for both dry and moist bare soil and vegetated conditions. Then the soil moisture of a pixel is estimated using the relative location of LST and f_C of that pixel within that trapezoid.

3.1.3 Output gap-filling

The CLMS ETa product consists of 5 sub-products: instantaneous sensible and latent heat fluxes (in W.m^{-2}), and dekadal (10-day mean) evapotranspiration and its components evaporation and transpiration (in mm.day^{-1}). The instantaneous heat fluxes represent values modelled at the time of Sentinel-3 satellite overpass and, therefore, do not undergo any gap-filling. On the other hand, the water fluxes need to undergo gap-filling. Otherwise, the dekadal aggregation would only take into account fluxes modelled during clear-sky conditions within the aggregation period. This would firstly result in frequent gaps and secondly in a systematic overestimation of the dekadal aggregate.

The gap-filling of ET product, i.e. estimation of ET during cloudy conditions, is usually performed using a reference quantity that can be derived for any date regardless of cloud conditions. This implies that this reference quantity is mostly dependent on weather forcing. In the Sen-ET approach the choice was made to use reference evapotranspiration calculated using the FAO-56 method (Allen, 1998). A ratio of modelled daily ETa to reference ET (called crop-stress coefficient $K_{s,c}$) is calculated on dates for which daily ETa is available and is used to recreate ETa on the target date which needs to be gap-filled, as described in Guzinski et al. (2021). This method was further developed in Guzinski et al. (2023) to better account for soil drying by performing linear

interpolation of $K_{s,c}$ (applicable only in non-Near-Real-Time (NRT) modelling when ETa after the target date is available) and for soil wetting through rainfall by setting $K_{s,c}$ to the maximum value observed during the gap-filling period if a simple water-balance approach indicates that the soil is wet. Taking rainfall into account is important especially for longer gap-filling periods (Delogu et al., 2012) and in climates in which rainfall initiates the growing season (e.g. rainy season in the Sahel and other semi-arid areas). For the CLMS ETa processing chain we further improved the robustness of this method, especially for longer gap-filling windows, by replacing the maximum $K_{s,c}$ observed within the gap-filling window with the 80th percentile of ratios observed within the gap-filling period. If the latest know $K_{s,c}$ before the target date was larger than 80th percentile then the value of that last know $K_{s,c}$ was preserved. This was done to ensure that taking rainfall into account would only increase, and not reduce, the gap-filled ET values. To accommodate longer periods with few satellite observations a 60-day gap-filling window is used during ETa production.

Once the gap-filled ET is estimated, the split into evaporation and transpiration is performed using the ratio of evaporation or transpiration to ET from the closest non-gap-filled preceding date.

3.1.4 Ensemble model

Once TSEB-PT and ETLook are run and gap-filling is done, ET, T and E are aggregated at dekadal timescales, producing three timesteps each month:

1. From 1st to 10th day of the month
2. From 11th to 20th day of the month
3. From 21st to the end of the month

The ensemble model is then computed, for each dekad, as the output average of TSEB-PT and ETLook (Eq. 8):

$$ET_{ENS} = \frac{ET_{ETLook} + ET_{TSEB}}{2}$$
$$E_{ENS} = \frac{E_{ETLook} + E_{TSEB}}{2} \quad \text{Eq. 8}$$
$$T_{ENS} = \frac{T_{ETLook} + T_{TSEB}}{2}$$

In addition, we can compute the uncertainty associated to the model(s) as the standard deviation (STD) between both model outputs. Since we have only two models, and without any a priori

knowledge of the probabilistic distribution, we need to assume that all possible outputs could be uniformly distributed between the outputs yielded by TSEB and ETLook. Therefore, the ensemble model uncertainty is computed for ET, E and T as (Eq. 9):

$$ET_{STD} = \frac{|ET_{ETLook} - ET_{TSEB}|}{2\sqrt{3}}$$
$$T_{STD} = \frac{|T_{ETLook} - T_{TSEB}|}{2\sqrt{3}} \quad \text{Eq. 9}$$
$$E_{STD} = \frac{|E_{ETLook} - E_{TSEB}|}{2\sqrt{3}}$$

3.1.5 Quality parameters

The dekadal ETA product comes with three quality parameters:

1. the number of valid modelled clear-sky (i.e. not gap-filled) ET values within each dekad (NOBS). Since under certain conditions TSEB-PT can converge to a valid ET while ETLook cannot, and vice versa, this information is based on the Ensemble product (i.e. minimum NOBS from the two models).
2. The gap-filing distance (GFD). It represents the average temporal distance (in days) between the gap-filled ETs within a given dekad and the closest previous valid clear-sky modelled ET. Since under certain conditions TSEB-PT can converge to a valid ET while ETLook cannot, and vice versa, this metadata layer is based on the Ensemble product (i.e. maximum GFD of both models).
3. Some flags, currently based on binary validity of model outputs (i.e. modelled or not modelled) and landcover map used during the modelling. The flag's encoding is shown in Table 1. Note that the flags refer to the Ensemble product, i.e. bit 0 could be set when there is no Ensemble ETA value in a given pixel but in some cases either TSEB-PT or ETLook could still have converged to a valid ETA in the same pixel.

The daily Heat Flux product comes with one quality parameter. The thermal image acquisition overpass time (OT) represents a timestamp at which thermal image was acquired on a given day. This is also the time for which the instantaneous H and LE heat fluxes are modelled by the TSEB-PT model.

Table 1: Annotation flag encodings.

Bit	Name	Value
0	ET not modelled – cloud, snow, offshore or model non-convergence	1
1	Inland open-water pixel – ET not modelled	2
2	Urban pixel – ET modelled but with very high uncertainty	4
3	Currently unused	8
4	Currently unused	16
5	Currently unused	32
6	Currently unused	64
7	Currently unused	128

3.2 THE RETRIEVAL ALGORITHM

3.2.1 Outline

The general processing workflow for deriving the input forcing required by both ETa models at 300m resolution are depicted in Figure 1.

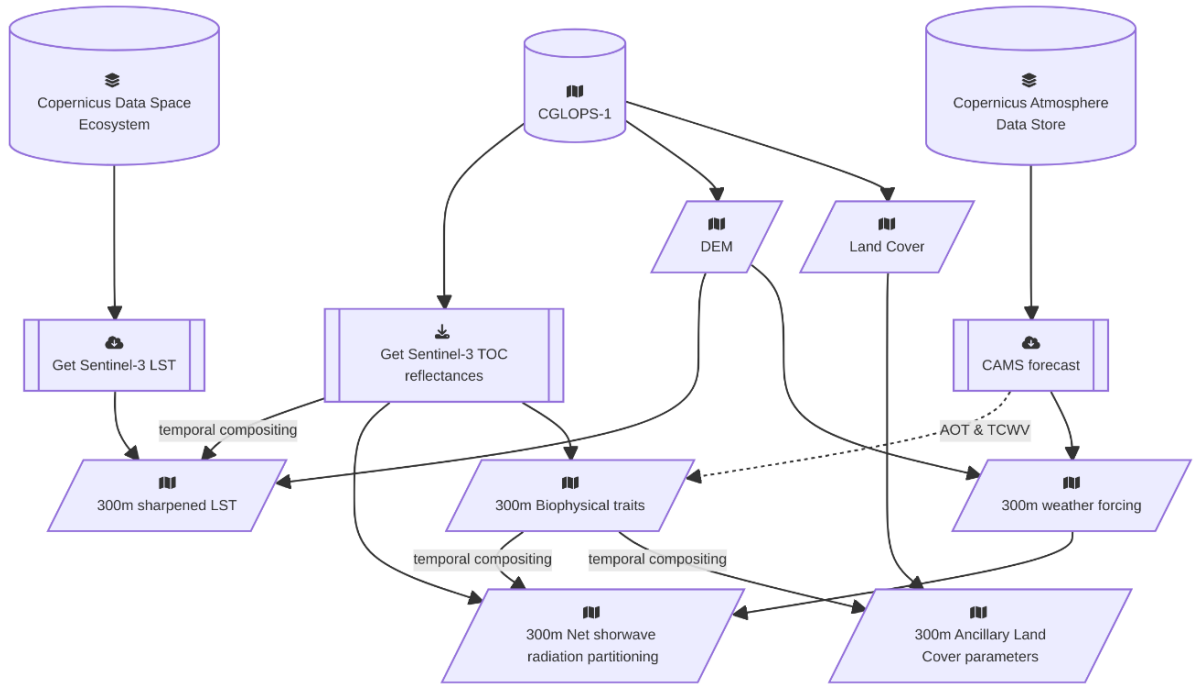


Figure 1: Processing workflow for the retrieval of input forcing for both TSEB-PT and ETLook models

3.2.2 Basic underlying assumptions

- The latent heat flux (or ET) is computed as residual from the surface energy balance, considering that heat advection, canopy heat storage and latent heat for carbon fixation are negligible.
- The calculation of the aerodynamic resistances in both ETLook and TSEB rely on the Monin-Obukhov similarity theory. Therefore, canopies at pixel level (300m) are homogeneous in terms of a single plant functional type.
- Gap-filling assumes a linear relationship between actual ET and FAO56 reference ET: $ET = k_{cs} ET_{ref}$ (see section 3.1.3 for more details)

3.2.3 Related and previous applications

- WaPOR: The water productivity portal of the Food and Agriculture Organisation (FAO) of the United Nations (https://www.fao.org/in-action/remote-sensing-for-water-productivity/en) provides dekadal ETA maps at three different scales, from local (30m) to global (300m). Currently, LST from the Visible Infrared Imaging Radiometer Suite (VIIRS) is used as main input thermal infrared (TIR) dataset
- SenET: a framework producing operational ET estimates using Copernicus data at 20m scale (https://eo4society.esa.int/projects/sentinels-for-evapotranspiration-sen-et/). It is the precursor of the input processing framework described in this ATBD.

3.2.4 Alternative methodologies currently in use

The WaPOR global ETa dataset is probably the closest alternative in terms of modelling framework (i.e. ETLook model) and spatial resolution of 300m. Apart from WaPOR (the long-term continuity of which is not guaranteed), global operational ETa datasets with closest matching spatio-temporal specifications are the MODIS and VIIRS ETa products (Román et al., 2024). They have a higher temporal resolution (8-days) but lower spatial resolution (500 m) and use a modeling approach which does not make direct use of LST measurements (Mu et al., 2011). Another operational and global product which utilizes MODIS and VIIRS data is produced by United States Geological Survey using SSEBop energy balance model (Senay et al., 2013) with dekadal temporal resolution and 1 km spatial resolution. Other ETa datasets have either much lower spatial and temporal resolutions (e.g. ETa product based on geostationary observations produced by the EUMETSAT Satellite Application Facility on Land Surface Analysis - (Barrios et al., 2024) at 5km resolution and microwave-based GLEAM at 0.1º), do not have global coverage (e.g. OpenET - (Melton et al., 2022)) or are not produced operationally and in NRT (e.g. ETMonitor - (Zheng et al., 2022)).

3.2.5 Input data

The CLMS ETA product is derived from the Sentinel-3 Top-Of Canopy (TOC) reflectances v2.3 [CGLOPS1_PUM_S3-TOC-Reflectance300m-V2.3] (Table 2). They are provided on a 10° x 10° tiles.

Table 2: List of input data fields used from the CLMS Sentinel-3 TOC reflectances

Name	Units	Description
Oa02_toc	–	TOC reflectance for OLCI band (412nm)
Oa03_toc	–	TOC reflectance for OLCI band (443nm)
Oa04_toc	–	TOC reflectance for OLCI band (491nm)
Oa05_toc	–	TOC reflectance for OLCI band (510nm)
Oa06_toc	–	TOC reflectance for OLCI band (560nm)
Oa07_toc	–	TOC reflectance for OLCI band (620nm)
Oa08_toc	–	TOC reflectance for OLCI band (665nm)
Oa09_toc	–	TOC reflectance for OLCI band (674nm)
Oa10_toc	–	TOC reflectance for OLCI band (681nm)
Oa11_toc	–	TOC reflectance for OLCI band (709nm)
Oa12_toc	–	TOC reflectance for OLCI band (754nm)
Oa16_toc	–	TOC reflectance for OLCI band (779nm)
Oa17_toc	–	TOC reflectance for OLCI band (865nm)
Oa18_toc	–	TOC reflectance for OLCI band (884nm)
Oa21_toc	–	TOC reflectance for OLCI band (1 013nm)
S5_an_toc	–	TOC reflectance for SLSTR band, nadir (1613nm)
S6_an_toc	–	TOC reflectance for SLSTR band, nadir (2255nm)
SAA_SLSTR		Solar azimuth angle for SLSTR
SZA_SLSTR		Solar zenith angle for SLSTR
VAA_SLSTR		Viewing azimuth angle for SLSTR
VZA_SLSTR		Viewing zenith angle for SLSTR
Cloud_an	–	Cloud mask from SLSTR level 1b input data
Quality_flags	–	Quality flags from OLCI level 1b input data
AC_process_flag	–	Quality flag of the atmospheric correction processing

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The LST products (SL_2_LST___), derived from the Sea and Land Surface Temperature Radiometer (SLSTR) onboard the Sentinel-3 platform, are retrieved from the Copernicus Data Space Ecosystem (CDSE) platform, from which we get the illumination and observation angles and the LST for clear-sky land pixels (Table 3).

The weather forcings are retrieved from the Copernicus Atmosphere Monitoring Service (CAMS) Data Store. For each day, we download the CAMS Global Atmospheric Composition Forecast (global-atmospheric-composition-forecasts) dataset in a 3-day window in order to allow the temporal daily integration globally (Table 4).

Finally, the Copernicus Digital Elevation Model (DEM) (COP-DEM_GLO-90-DTED), the CLMS global annual land cover (https://land.copernicus.eu/en/products/global-dynamic-land-cover) and a Global Ecosystem Dynamics Investigation (GEDI) global canopy height (Lang et al., 2023) are used to produce the static ancillary inputs. These products are resampled at 300m using average resampling and create tiles of 10° x 10.

Table 3: List of input data fields used from the Sentinel-3 LST product

Dataset	Name	Units	Description
LST_in.nc	LST	K	Land surface temperature
geometry_tn.nc	sat_zenith_tn		Satellite zenith viewing angle
	sat_azimuth_tn		Satellite azimuth viewing angle
	solar_zenith_tn		Solar zenith angle
	sat_azimuth_tn		Solar azimuth angle
flags_in.nc	cloud_in	–	Threshold based cloud test
	bayes_in	–	Probabilistic cloud mask

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Table 4: List of CAMS global atmospheric composition forecasts downloaded in the physical units provided by the CAMS data store

Variable	Unit	'cdsapi' name
10m u-component of wind	m s^{-1}	10m_u_component_of_wind
10m v-component of wind	m s^{-1}	10m_v_component_of_wind
2m dewpoint temperature	K	2m_dewpoint_temperature
2m temperature	K	2m_temperature
Geopotential height	m^{-2} s^{-2}	surface_geopotential
Surface pressure	Pa	surface_pressure
Total aerosol optical depth at 550 nm	–	total_aerosol_optical_depth_550 nm
Total column water vapour	kg m^{-1}	total_column_water_vapour
Surface solar radiation downwards	J m^{-2}	surface_solar_radiation_downwards
Surface thermal radiation downwards	J m^{-2}	surface_thermal_radiation_downwards
Total precipitation	m	total_precipitation

3.2.6 Output product

The dekadal ETA output product consists in 15 variables while the daily Heat Flux output product consists of 3 variables as shown in Table 5.

Table 5: Content of the ETA and Hea Flux output products.

Product	Variable	Description
ETA	ET-TSEBPT	actual evapotranspiration estimated by the TSEB-PT model
	ET-ETLOOK	actual evapotranspiration estimated by the ETLook model
	ET-ENSEMBLE	actual evapotranspiration estimated by the Ensemble model
	ET-STD	Standard deviation (uncertainty) of ET estimate by the Ensemble model
	E-TSEBPT	Soil evaporation estimated by the TSEB-PT model
	E-ETLOOK	Soil evaporation estimated by the ETLook model
	E-ENSEMBLE	Soil evaporation estimated by the Ensemble model
	E-STD	Standard deviation (uncertainty) of E estimate by the Ensemble model
	T-TSEBPT	Canopy transpiration estimated by the TSEB-PT model
	T-ETLOOK	Canopy transpiration estimated by the ETLook model
	T-ENSEMBLE	Canopy transpiration estimated by the Ensemble model
	T-STD	Standard deviation (uncertainty) of T estimate by the Ensemble model
	NOBS	Number of cloud free observations in a given dekad in Ensemble model output
	GFD	Average number of days to closest non-cloudy observation for gap-filled dates in Ensemble model output
	FLAG	Annotation flags for Ensemble model output
Heat Flux	H-TSEBPT	Instantaneous sensible heat flux estimated by the TSEB-PT model at the time of satellite observation
	LE-TSEBPT	Instantaneous latent heat flux estimated by the TSEB-PT model at the time of satellite observation
	OT	Overpass time (UTC) of the satellite observation

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3.2.7 Detailed algorithmic description

3.2.7.1 Weather forcing

Weather forcing is critical for accurate estimation of ET. Due to the 2-day timeliness requirements for the Near-Real-Time (NRT) product, the weather data source is Copernicus Atmosphere Monitoring Service (CAMS) forecasts, produced by the European Center for Medium Range Weather Forecasts and distributed freely and openly through the CAMS Data Store (https://atmosphere.copernicus.eu/data). CAMS data contain surface meteorological parameters covering the whole Earth on a 0.4° grid resolution and hourly temporal resolution.

Instantaneous weather forcing at the satellite overpass are used to drive the ET models and include air temperature, vapor pressure, wind speed, surface pressure, and clear-sky solar irradiance. All instantaneous data were obtained by linear interpolation between two CAMS hourly forecasts to the time of Sentinel-3 SLSTR acquisition over the area of interest/tile. Daily weather forcing is used to extrapolate and interpolate the instantaneous estimates of ET and include solar irradiance as well as air and dew temperatures, wind speed, and pressure, which are then used to calculate the FAO-56 reference ET required for the gap filling (section 3.1.3). They are integrated over a 24-h period starting at midnight local time.

Both instantaneous and daily forcing are corrected for topography (Guzinski et al. 2021). To that purpose, the Copernicus DEM is resampled at a resolution of 300 m to enhance the spatial resolution of the meteorological inputs as described below and depicted in Figure 2.

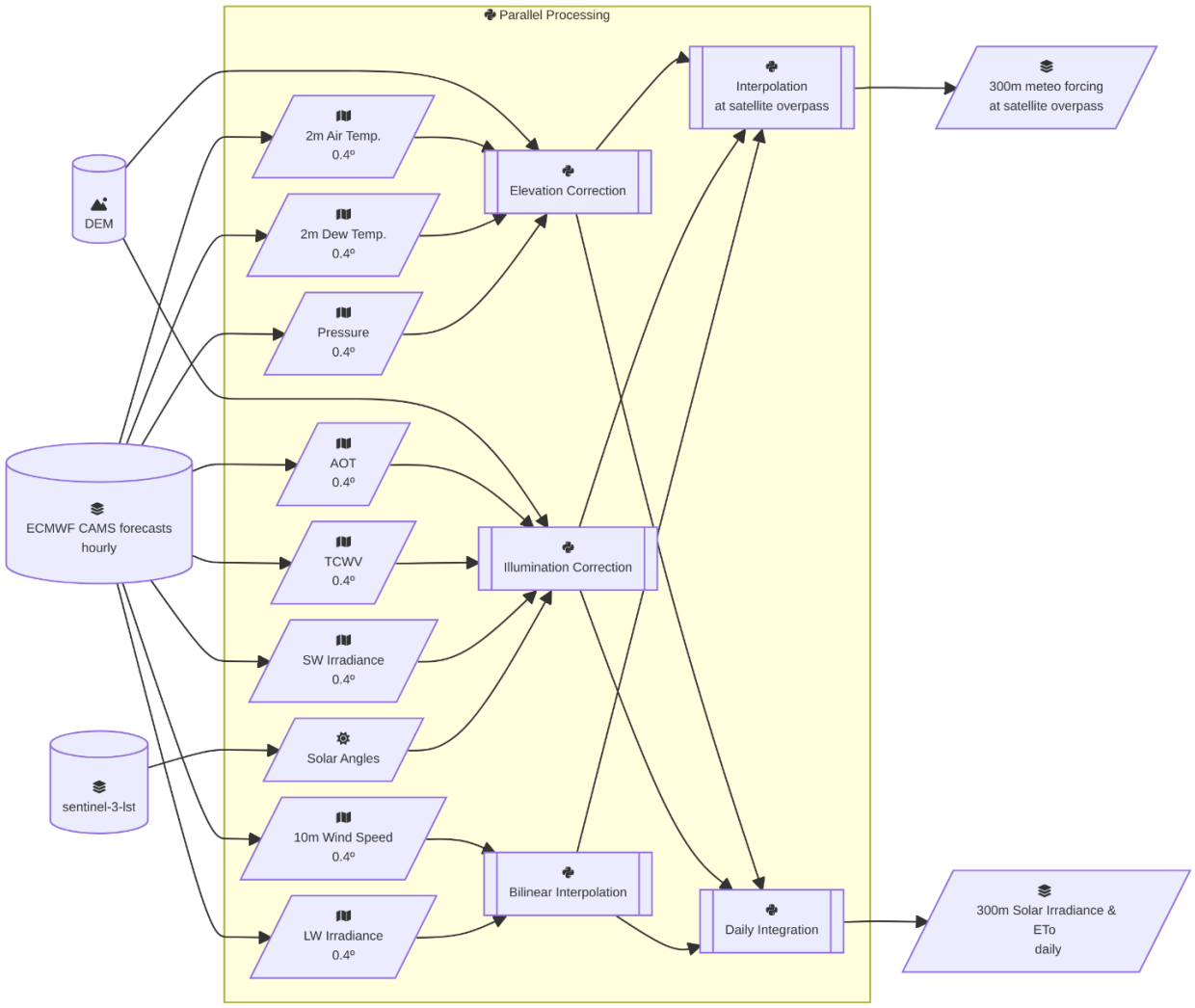


Figure 2: Weather forcing processing workflow applying the topographic corrections

- Instantaneous air temperature is based on 2-m air temperature CAMS variable. Due to the low spatial resolution of the meteorological data, it is assumed that the air temperature that better represents meteorological conditions at that resolution is the temperature at atmospheric blending height (set to be 100 m above ground), where the impact of local surface conditions on those parameters is not so direct. The resolution of air temperature is additionally enhanced by using the 300-m resolution DEM and a standard lapse rate of 0.0065 K m^{-1} (Sissenwine et al., 1962) to correct for temperature variations due to changes

in elevation between the 2 m above the geopotential height at which CAMS temperature is produced and the 100 m above DEM surface elevation.

- Vapor pressure is derived from 2-m dew point temperature CAMS field. Similarly to air temperature, it is assumed to represent conditions at blending height and is corrected for changes in elevation using DEM and moist air lapse rate of 0.002 K m^{-1} (Sissenwine et al., 1962).
- Surface pressure, based on CAMS field of the same name, is corrected for changes due to varying elevation using a DEM.
- Surface shortwave irradiance is computed assuming clear sky conditions, since this parameter is only used at a time and place where thermal observations of the surface by the Sentinel-3 satellite were possible. Its calculation is based on the REST2 model (Gueymard, 2008), using the Aerosol Optical Thickness at 550nm and the Total Column Water Vapor of CAMS. In addition, solar irradiance is also corrected by elevation, incidence angle, and terrain shading based on the DEM information on elevation, slope and terrain orientation to estimate irradiance on a tilted surface (Aguilar et al., 2010).
- Wind speed is based on 10 m “u” and “v” CAMS wind components. Apart from trigonometric calculation to obtain total magnitude of wind speed, no other preprocessing is performed.
- Daily solar irradiance is calculated using 24-h integration of CAMS downward surface irradiance product and clear sky irradiance estimated as described previously. At each hourly timestep, a cloudiness factor is first estimated using the ratio of surface to horizontal clear sky irradiance. Based on this, beam and diffuse radiation components are derived and corrected for terrain orientation similarly as is done with the instantaneous clear-sky irradiance (Aguilar et al., 2010) to finally integrate these hourly values to daily.
- The daily reference ET is estimated using the FAO56 model (Allen, 1998). It is calculated using daily mean values of air temperature, dew point temperature, wind speed, surface pressure, and surface solar radiation products. All the relevant terrain corrections are applied to those products before they are used in the FAO56 model.
- Daily total precipitation is calculated using 24-h integration of CAMS hourly total precipitation forecasts. Similarly, as with wind speed, no other preprocessing is performed, due to the complexity/impossibility of downscaling such products.

3.2.7.2 Biophysical traits

Besides LST and weather forcing, some biophysical traits are mainly needed to evaluate the amount of radiation that is intercepted and absorbed by the canopy. Therefore, LAI is key in ETA modelling since it is the main driver, together with the leaf inclination distribution function, in partitioning the radiation that is intercepted by the canopy or transmitted towards the soil. Indeed, all plant materials intercept radiation and thus they contribute to the amount of energy that the canopy has available to partition the heat between the latent heat flux (i.e. transpiration) and the sensible heat flux (i.e. increase the surface temperature). On the other hand, only the green leaves are able to transpire through the stoma opening. For that reason, several ETA models, including TSEB, require an estimate of both the total LAI (or more precisely Plant Area Index in the case of woody vegetation) and the photosynthetically active LAI (or green LAI). Furthermore, leaf pigments might also come useful in order to derive leaf albedo, as we will see in section 3.2.7.3.3 later on.

A sound and generalizable approach to obtain the biophysical traits required by the ETA models is the inversion of canopy radiative transfer models (RTM), such as the combined ProspectD (Féret et al., 2017) and 4SAIL (Verhoef et al., 2007) RTMs. Prospect (in its different versions) together with 4SAIL RTM can simulate the canopy spectra from 400 to 2500 nm at every nm based on the inputs listed in Table 6.

Several algorithms have been developed to operationally invert these RTMs in order to obtain (some of) the biophysical traits of Table 6. One of the sound approaches, in terms of computing cost efficiency, consists on a hybrid inversion, in which a regression model is trained based on large number of Prospect and SAIL (in their different model versions) simulations that have previously been computed (Weiss et al., 2002, 2000). Then this regression model is applied to the spectral imagery to obtain the biophysical traits maps.

Table 6: List of biophysical traits and ancillary information required by ProspectD+4SAIL radiative transfer model

Type	Variable	Units
Leaf trait	Leaf structure parameter	–
	Chlorophyll a+b concentration	µg cm^{-2}
	Carotenoids concentration	µg cm^{-2}
	Anthocyanins concentration	µg cm^{-2}
	Brown pigments	arbitrary
	Dry matter content	g cm^{-2}
	Water content	g cm^{-2}
Canopy trait	Total LAI	m^2 m^{-2}
	LIDF: Campbell (1990) mean leaf angle	°
	hotspot parameter	–
Ancillary inputs	Soil spectrum	–
	direct/diffuse irradiance	–
	solar angles	°
	sensor angles	°

In this case, we use our own vectorized version of ProspectD and 4SAIL models, available at https://github.com/hectornieto/pypro4sail. This vectorized version allows us to run very efficiently the RTMs and thus enable us to generate specific regression models for each scene, given the actual observation and illumination conditions of each scene. This contrasts with other similar algorithms, such as the BiophysicalOp in the Sentinel-2 toolbox [ATBD_S2ToolBox], which uses a generic regression model for all scenes instead. The processing framework depicted in Figure 3 is summarized as:

- Mean sun zenith and azimuth angles together with mean aerosol optical thickness and total column water vapor at the scene are used as inputs for 6S model (Vermote et al., 1997) in order to get the proportion of spectral direct and diffuse irradiance (SKYL in Figure 3).
- We generate ca. 40000 ProspectD+4SAIL simulated spectra by:

- creating the same amount of Monte Carlo random samples (Saltelli et al., 1999) of biophysical traits and observation angles, based on prescribed plausible ranges listed in Table 7.

Table 7: Range of values that are simulated. The units of each trait are shown in Table 6. Ranges are based on information extracted from the LOPEX database (Hosgood et al., 2005) and the Sentinel-2 Biophysical ATBD [ATBD_S2ToolBox]

Variable	Lower range	Upper range
Leaf structure parameter	1	3
Chlorophyll a+b concentration	0	110
Carotenoids concentration	0	30
Anthocyanins concentration	0	40
Brown pigments	0	2
Dry matter content	0.0017	0.0031
Water content	0	0.0525
Total LAI	0	8
Campbell (1990) mean leaf angle	20	80
hotspot parameter	0.05	1
View Zenith Angle	Min. angle of the scene	Max. angle of the scene
Relative Sun-View Azimuth Angle	Min. angle of the scene	Max. angle of the scene
soil spectrum	ECOSTRESS spectra library	

- running ProspectD+4SAIL for each of these samples together with the generated proportion of spectral direct and diffuse irradiance and the mean sensor and solar angles at the scene.
- Convolving the simulated narrowband spectra using the Sentinel OLCI & SLSTR spectral response functions in order to obtain a set of 40000 simulated Sentinel-3 TOC reflectances.

- Train a random forest regression in which the dependent variables are the randomly generated biophysical traits and the explicative variables the correspondent simulated TOC reflectances.

- Apply the random forest regression to the actual TOC reflectance scene.

As outputs, we obtain 8 products: the LAI, the Campbell (1990) mean leaf angle, the leaf chlorophyll a+b concentration, the leaf carotenoids concentration, the leaf anthocyanins concentration, the brown pigments, the leaf dry matter content, and the leaf water content. In order to test the goodness of the random forest inversion and the uncertainty of such inversion method, we run an independent set of ProspectD+4SAIL simulations to compare how the regression model predicts the biophysical traits. This is shown as an example in Figure 4.

Finally, the retrieved leaf pigments concentration gives us an idea about the canopy greenness, and indeed we have observed an increase of Sentinel-3 estimated anthocyanins concentration during vegetation curing in summer and the leaf senescence in fall: anthocyanins are red pigments and thus may become dominant over other leaf pigments in those situations (Féret et al., 2017). For that reason, we can express the fraction of LAI that is green (f_g) using an empirical piecewise linear relation to the anthocyanins (Ant) (Eq. 10):

$$f_g = \begin{cases} 1, & \text{if } Ant \leq 5 \mu g cm^{-2} \\ 1 - 0.8\dfrac{Ant - 5}{20}, & \text{if } Ant > 5 \mu g cm^{-2} \text{ and } Ant \leq 25 \mu g cm^{-2} \\ 0.2, & \text{if } Ant > 25 \mu g cm^{-2} \end{cases} \quad \text{Eq. 10}$$

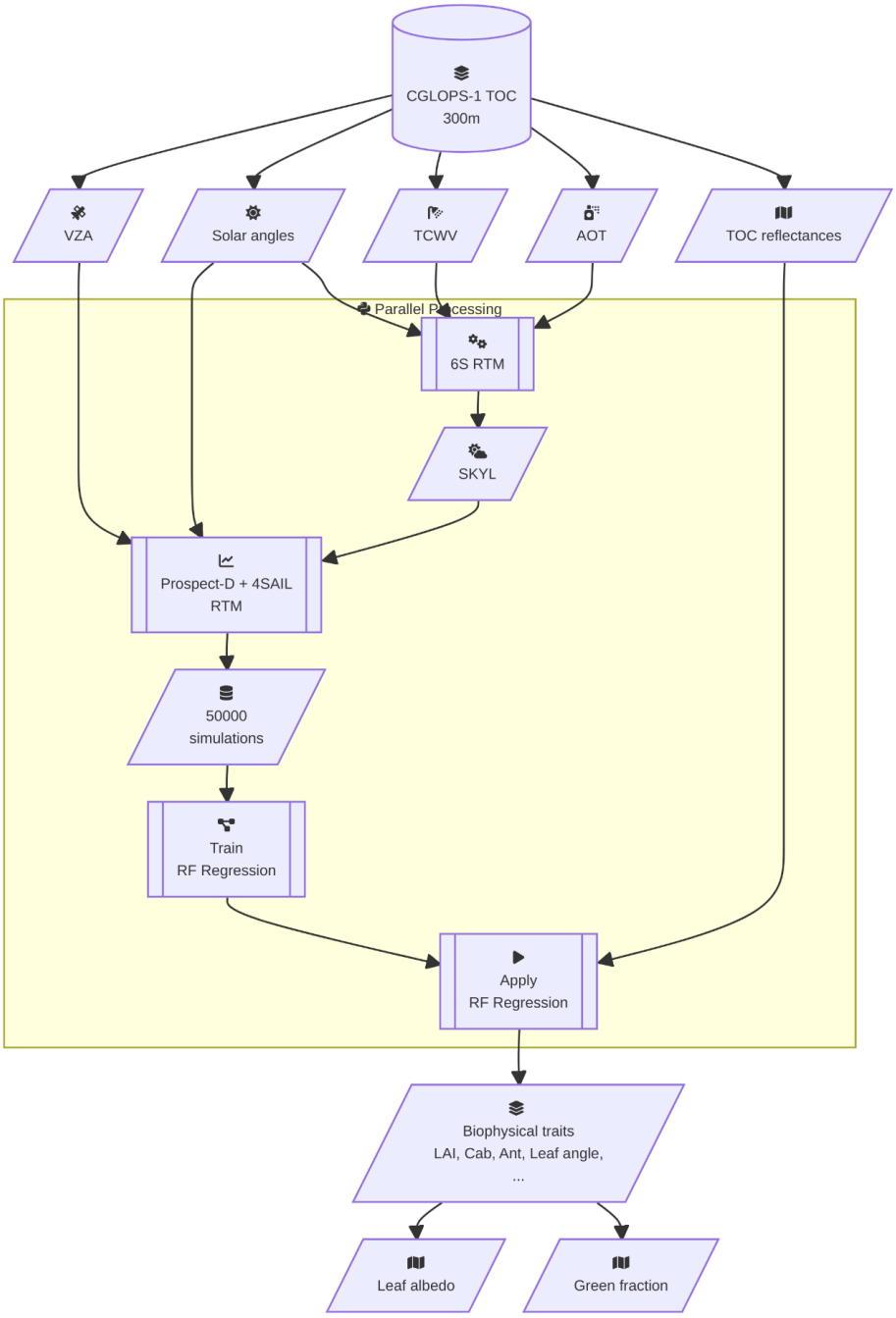


Figure 3: Biophysical traits processing workflow using the ProspectD+4SAIL hybrid inversion setup

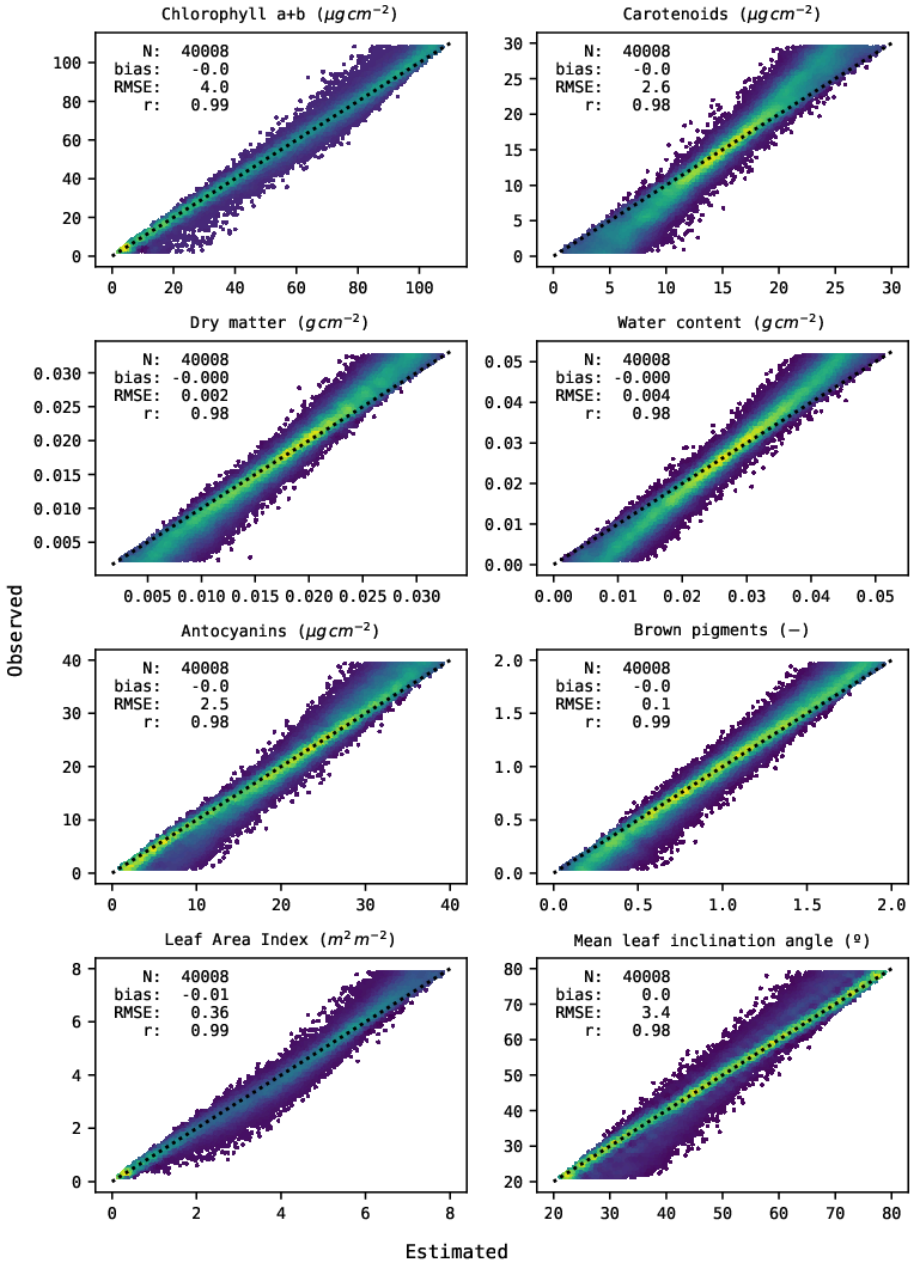


Figure 4: Evaluation performance for the Random Forest hybrid inversion of the ProspectD+4SAIL radiative transfer model. The "observed" dataset corresponds to ca. 40000 independent simulation of ProspectD+4SAIL for a VZA=0°, SZA=37.5° and a standard atmosphere

3.2.7.3 Albedo / net shortwave radiation

The net radiation (R_n) is computed as (Eq. 11):

$$R_n = S^{\downarrow}(1-\alpha) + \epsilon\left(L^{\downarrow} - \sigma LST^4\right) \quad \text{Eq. 11}$$

where $S^{\downarrow}$ and $L^{\downarrow}$ are the shortwave and longwave incoming irradiances, respectively. α and ϵ are surface albedo and emissivity, respectively, LST is the Land Surface Temperature, and $\sigma \approx 5.67 \cdot 10^{-8}$ W m^{-2} K^{-4} is the Stefan-Boltzmann constant. α and ϵ (as well as LST) can be estimated, more or less accurately, from Earth Observation (EO) data.

Considering the larger magnitude of shortwave irradiance ($S^{\downarrow}$) compared to the longwave irradiance ($L^{\downarrow}$), and the fact that L_n is usually computed internally by each ET model, we focus on the shortwave net radiation (S_n). In particular, we concentrate on the canopy and leaf properties that influence albedo and radiation partitioning between soil and canopy.

The albedo (α) is the key variable that determines the amount of intercepted light that is absorbed by the surface. α is defined as the proportion of incident shortwave radiation that is reflected by the surface. The shortwave net radiation (S_n) is therefore the balance between the incident shortwave irradiance ($S^{\downarrow}$) and the reflected shortwave radiance ($S^{\uparrow} = \alpha S^{\downarrow}$) (Eq. 12)

$$S_n = S^{\downarrow}(1-\alpha) \quad \text{Eq. 12}$$

The spectral properties of the surface are key in determining the albedo. For instance, fresh snow shows large values ($\alpha \approx 0.9$) and, thus, reflects most of the solar irradiance, while oceans have much lower values ($\alpha \approx 0.05$), absorbing a large amount of solar irradiance. Leaves, due to their photosynthetic activity, absorb a large proportion of light due to the presence of chlorophylls, as well as other leaf pigments. For that reason, leaves yield relative low albedo values (≈ 0.15). On the other hand, soils can have a large range of albedo values, depending on their mineral composition, texture and topsoil moisture. Therefore, the albedo of a vegetated surface will depend not only on leaf chlorophyll concentration but also on canopy density and, in a lesser degree, on the soil albedo in situations of sparse vegetation or initial growth stages.

3.2.7.3.1 Narrowband2broadband processor

The first step in computing the net shortwave radiation is to convert the narrowband TOC reflectances into broadband reflectances. For that purpose, we apply the approach proposed by Liang (2001), who derived a linear regression model between the satellite spectral bands and the broad bands corresponding to the PAR region (400–700 nm), the NIR (700–2500 nm) and the solar spectrum (400–2500 nm). Similarly as Liang (2001), in order to overcome the limited amount of *in situ* measurements of albedo, we made use again of the ProspectD+4SAIL canopy RTM coupled with the 6S atmospheric RTM to run a large number of simulations covering all plausible illumination, atmospheric and canopy conditions. Once the RTM simulations are performed (at 1 nm step), we convolved to both Sentinel-3A and Sentinel-3B specific spectral response functions, and we also integrated these simulated spectra to the broadband regions defined above. Then, one multivariate linear regression model per broadband region and per platform is fitted with the broadband reflectances as dependent variables and the Sentinel-3 simulated TOC reflectances as explanatory variables. The resulting coefficients are shown in Table 8, showing the weight of each Sentinel-3 band has on the broadband reflectances. It is worth noting that the coefficients are consistent with the expected behavior, since the spectral bands located in the visible has no influence on the NIR broadband reflectances, nor the spectral bands in the NIR/SWIR influence on the PAR broadband model.

Table 8: Narrowband to Broadband conversion coefficients for Sentinel-3A and Sentinel-3B platforms

	BAND	O2	O3	O4	O5	O6	O7	O8	O9	O10	O11	O12	O16	O17	O18	O21	S5	S6
3A	SW	0.10	-0.18	0.23	0.09	0.08	0.03	0.13	-0.41	0.32	0.04	0.09	0.01	0.21	-0.13	0.19	0.12	0.02
	PAR	0.10	0.14	-0.01	0.24	0.17	0.24	0.15	-0.26	0.21	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	NIR	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.04	0.17	0.00	0.55	-0.39	0.40	0.19	0.07
3B	SW	0.10	-0.18	0.23	0.09	0.08	0.03	0.14	-0.42	0.32	0.04	0.09	0.01	0.21	-0.13	0.19	0.12	0.02
	PAR	0.10	0.14	-0.01	0.25	0.17	0.24	0.16	-0.28	0.21	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	NIR	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.04	0.17	0.00	0.55	-0.38	0.40	0.19	0.07

Furthermore, we evaluated the goodness of this approach by running an independent set of simulations and compared the retrieved broadbands using the coefficients of Table 8 (Figure 5). For both platforms, the evaluation confirms the robustness in the conversion from the Sentinel-3 TOC reflectances to the broadband PAR, NIR and SW spectral regions, with negligible mean bias (~0), very low RMSE and very good correlation (~1).

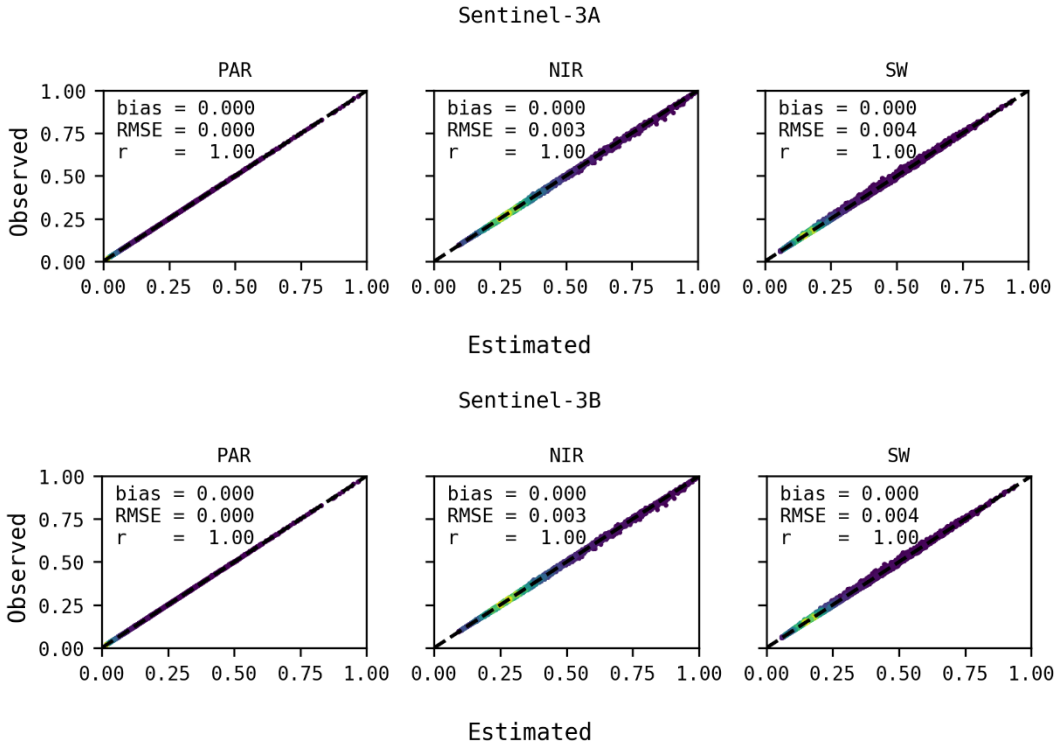


Figure 5: Evaluation of the narrowband to broadband conversion for the estimated Sentinel-3A (top) and Sentinel-3B (bottom) coefficients using the Liang (2001) method. These results are obtained after applying the coefficients in Table 8 to 40000 ProspectD+4SAIL independent simulations.

3.2.7.3.2 Campbell RTM

To compute the net shortwave radiation and its partitioning between the canopy and the soil we are using the model of Campbell and Norman (1998). The processing workflow is summarized in Figure 6.

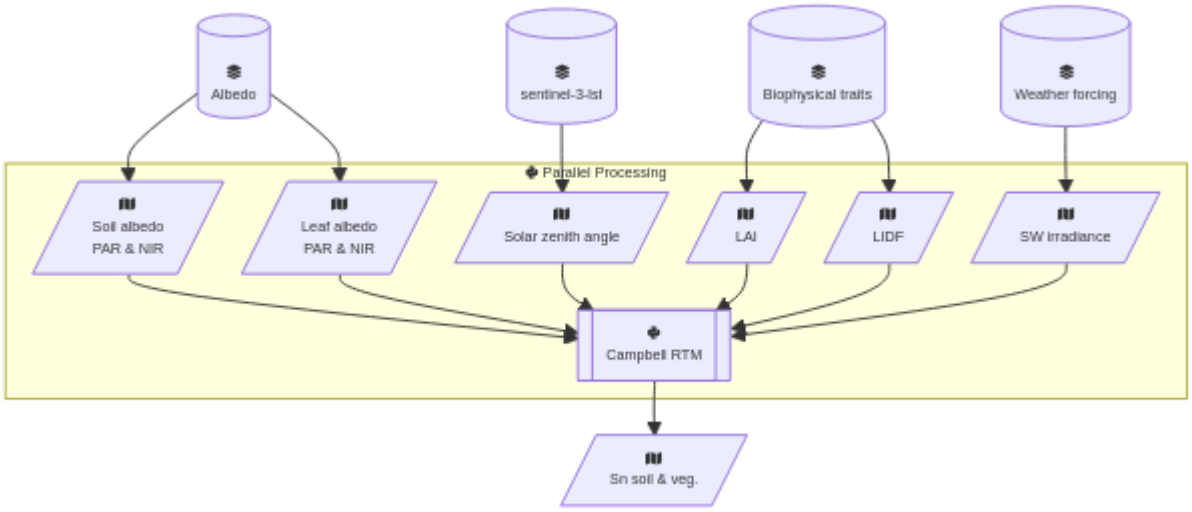


Figure 6: Net shortwave radiation processing workflow using the Campbell and Norman (1998) RTM-setup

The key aspect of this model is the calculation of the transmitted shortwave radiation through the canopy (τ_C). The transmittance of shortwave radiation through the canopy depends on the wavelength due to vegetation absorbing a greater portion of the photosynthetically active radiation (PAR, 400–700 nm spectrum) than near-infrared radiation (NIR, 700–2500 nm spectrum) wavelengths. (τ_C is partitioned into two components (direct/diffuse) and in two spectral band (PAR/NIR).

The direct-beam spectral transmittance ($\tau_{C,DIR,\lambda}$) at a given solar zenith angle (θ_s) is calculated following the equations of Campbell and Norman (1998) for a single layer crop (Eq. 13):

$$\tau_{C,DIR,\lambda}(\theta_s) = \frac{\left(\rho^*_{C,\lambda}(\theta_s)^2 - 1\right)exp\left(-\sqrt{\zeta_\lambda}\kappa_b(\theta_s)LAI\right)}{\left(\rho^*_{C,\lambda}\rho_{S,\lambda} - 1\right)+\rho^*_{C,\lambda}(\psi)\left(\rho^*_{C\lambda}(\theta_s) - \rho_{S,\lambda}\right)exp\left(-2\sqrt{\zeta_\lambda\kappa_b(\theta_s)LAI}\right)} \quad \text{Eq. 13}$$

with (λ) being either the PAR or NIR. $\rho^*_{C,\lambda}(\theta_s)$ is the beam spectral reflection coefficient for a deep canopy with non-horizontal leaves (see Eq. 14), ζ_λ is the leaf absorptance, κ_b is the extinction

coefficient for direct-beam radiation (per LAI unit), and $\rho_{S,\lambda}$ is the soil spectral reflectance. The multiple scattering between the soil and the canopy is accounted for in the $\rho^*_{C,\lambda}$ and $\rho_{S,\lambda}$ terms.

$$\rho^*_{C,\lambda}(\theta_s) = \frac{2\kappa_b(\theta_s)\rho^H_\lambda}{\kappa_b(\theta_s)+1} \quad \text{Eq. 14}$$

where $\rho^H_\lambda = \frac{1-\sqrt{\zeta_\lambda}}{1+\sqrt{\zeta_\lambda}}$ is the reflectance factor for a canopy with horizontal leaves.

Finally, the canopy beam extinction $\kappa_b(\psi)$ is calculated based on the ellipsoidal LIDF of Campbell (1990) (Eq. 15):

$$\kappa_b(\theta_s) = \frac{\sqrt{\chi^2+tan^2\theta_s}}{\chi+1.774(\chi+1.182)^{-0.733}} \quad \text{Eq. 15}$$

The diffuse spectral transmittance ($\tau_{C,DIF,\lambda}$) is calculated by numerically integrating κ_b over the hemisphere:

$$\kappa_d = 2\int_0^\pi \kappa_b(\psi) sin\psi cos\psi d\psi \quad \text{Eq. 16}$$

and replacing κ_b by κ_d in Eq. 15.

Similarly, the canopy direct spectral albedo is computed as (Eq. 17):

$$\rho_{C,DIR,\lambda}(\theta_s) = \frac{\rho^*_{C,\lambda}(\theta_s)+\left[\frac{\rho^*_{C,\lambda}(\theta_s)-\rho_{S,\lambda}}{\rho^*_{C,\lambda}(\theta_s)\rho_{s,\lambda}-1}\right] exp\left(-2\sqrt{\zeta_\lambda}\kappa_b(\theta_s)LAI\right)}{1+\rho^*_{C,\lambda}(\theta_s)+\left[\frac{\rho^*_{C,\lambda}(\theta_s)-\rho_{s,\lambda}}{\rho^*_{C,\lambda}(\theta_s)\rho_{S,\lambda}-1}\right] exp\left(-2\sqrt{\zeta_\lambda}\kappa_b(\theta_s)LAI\right)} \quad \text{Eq. 17}$$

and the diffuse canopy albedo ($\rho_{C,DIF,\lambda}$) by replacing $\kappa_b(\theta_s)$ by κ_d

Therefore, as inputs for computing net shortwave radiation we need to provide as input forcing the solar irradiance, direct and diffuse in the PAR and NIR, and then as main input the *LAI*, followed by soil and leaf spectra and the Campbell (1990) LIDF parameter ($\chi = \left(\frac{\text{leaf-angle}_{mean}}{9.65}\right)^{-1/1.65} - 3$).

3.2.7.3.3 Leaf bi-hemispherical reflectance and transmittance

For estimating leaf spectra, we use the leaf traits retrievals from the biophysical processor of section 3.2.7.2. With the information of these traits, we ran in forward mode the ProspectD leaf RTM to get a spectral reflectance and transmittance that could then be integrated to the required broadband regions. However, running ProspectD for a large array of pixels is too computationally expensive. For that reason, we have developed a ProspectD emulator: we generated broadband leaf reflectances and transmittances in the PAR and NIR from a large range of ProspectD simulations, covering all plausible range of leaf traits (see Table 7), which then is used to train another random forest model that relates the leaf traits to the broadband PAR and NIR leaf reflectance and transmittance. Figure 7 depicts the importance that each leaf trait has on the broadband reflectance emulator, showing that the most important pigment is the chlorophyll a+b concentration (Cab), due to its strong absorption of PAR radiation, followed by the leaf dry matter (Cm) and water (Cw) contents, which are the main absorbers of the NIR/SWIR radiation.

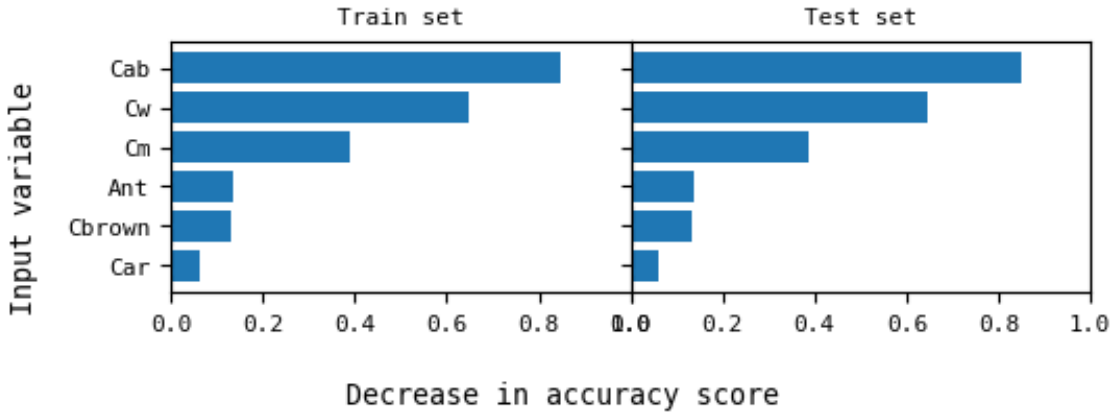


Figure 7: Mean permutation importance for lead biochemical components on the Prospect D leaf radiative transfer emulator.

On the other hand, the evaluation of the performance of the ProspectD emulator is shown in Figure 8.

Note that there is a larger scatter for the NIR reflectances and transmittances, likely due to the fact that we are intentionally excluding the ProspectD leaf structural parameter, that basically controls the multiple scattering within the leaf tissues, which is of a larger magnitude in the NIR region than

in the PAR due to smaller absorption coefficients in the NIR. However, the uncertainties when deriving the reflectances and transmittances seem to be cancelled out when computing the leaf absorptance (ζ=1 − ρ − τ). Since Campbell RTM only requires the leaf absorptance as input, the conversion from pigments to leaf spectra seems robust enough.

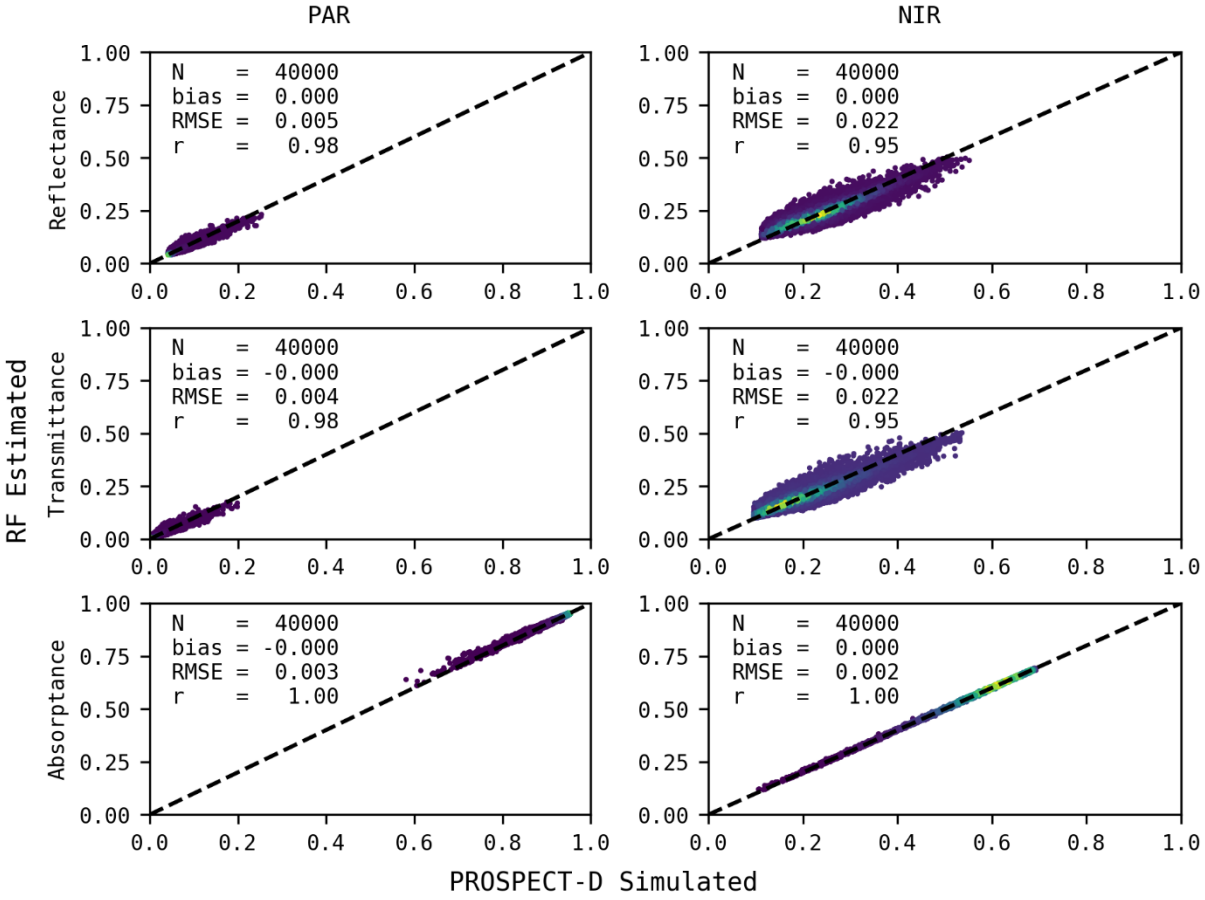


Figure 8: Emulator of ProspectD leaf radiative transfer model for the retrieval of broadband leaf reflectance, transmittance and absorptance factors.

3.2.7.3.4 Soil bi-hemispherical reflectance

Soil reflectance has a smaller role in most situations, since the radiation reaching the ground is smaller due to the interception of light by the canopy. However, in sparse and semi-arid conditions, where vegetation is scarce or even not present, soil albedo plays obviously a significant role.

Furthermore, these semi-arid areas are usually characterized by brighter soils since they are composed by sands, salts and with a small fraction of organic matter. For that reason, Guzinski et al. (2021) developed a method to unmix the broadband soil reflectance ($\rho_{soil,\lambda}$) from the TOC reflectance (Eq. 18)

$$\rho_{soil,\lambda}=\frac{\rho_{surface,\lambda}-[1-P_0(\theta_v)]\rho_{leaf,\lambda}}{P_0(\theta_v)} \quad \text{Eq. 18}$$

where $P_0(\theta_v)$ is the canopy gap fraction at the satellite observation angle, computed by the Beer-Lambert law ($exp[-\kappa(\theta_v)LAI]$). $\rho_{surface,\lambda}$ is the broadband surface reflectance, computed in Section 3.2.7.3.1, and $\rho_{leaf,\lambda}$ is the leaf broadband reflectance computed in Section 3.2.7.3.3.

To conclude, we summarize in Figure 9 the whole processing of obtaining the inputs required by the Campbell and Norman (1998) RTM.

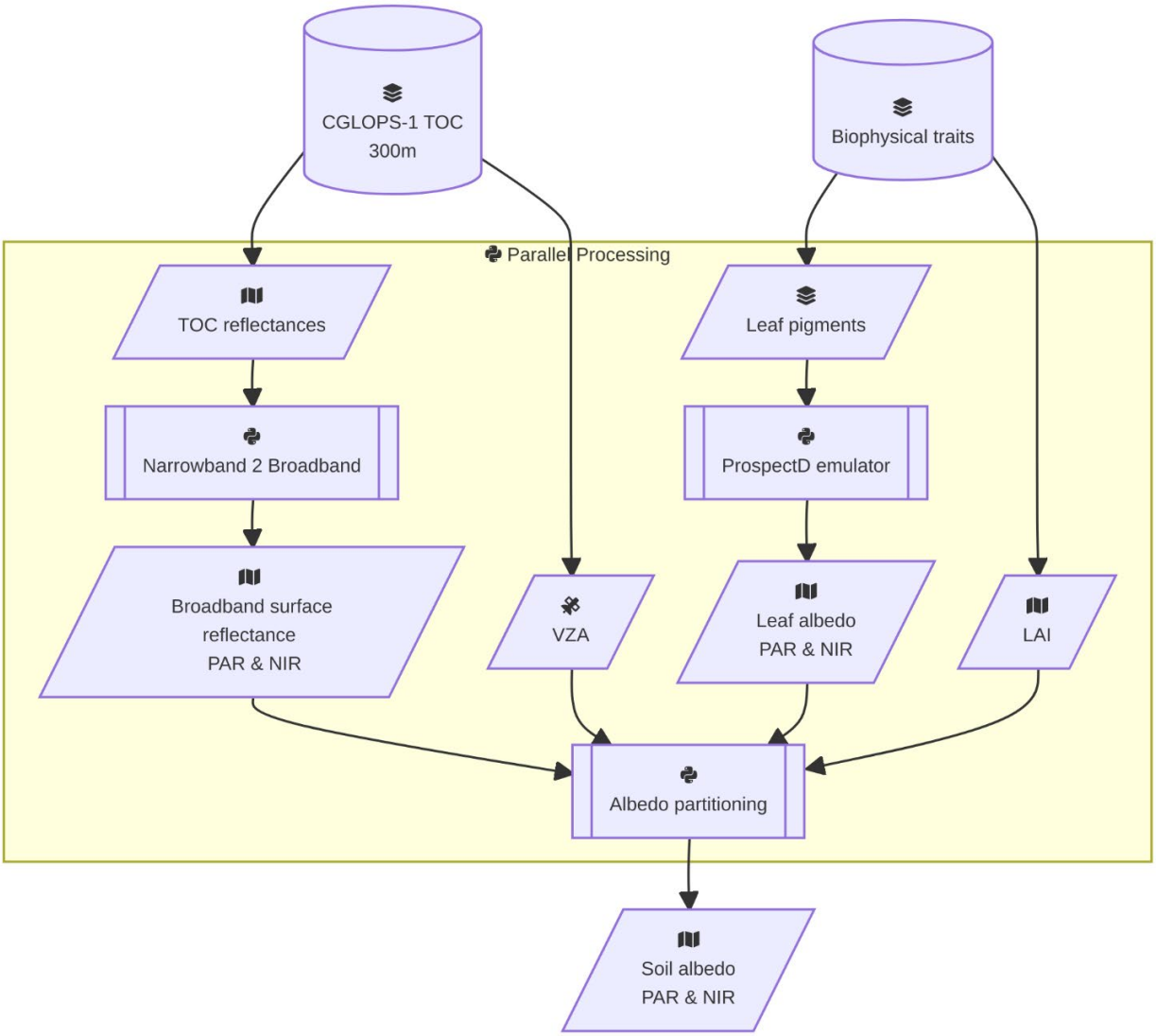


Figure 9: Leaf and soil albedo processing workflow under the Campbell and Norman (1998) RTM-based set-up

3.2.7.4 LST sharpening

The ETA product specification states a spatial resolution of 300 m. However, the spatial resolution of the SLSTR LST product is 1 km. Therefore, a method to perform thermal sharpening of the LST product is required. Most such methods are based on machine learning approaches, of various complexities, which capture the relationships between shortwave spectral reflectance (and possibly

other ancillary data) and the LST. The shortwave reflectance is of higher spatial resolution and is resampled to the resolution of LST to be used as explanatory variables for model training. Once trained, the model is then applied to the reflectance at its original resolution to obtain a representation of LST at that resolution. In most approaches, this is followed by a bias correction step to ensure conservation of energy between the LST maps at both the original and sharpened spatial resolutions.

Both the Sen-ET and WaPOR (version 3) modelling frameworks use a Data Mining Sharpener (DMS) thermal sharpening approach (Gao et al., 2012) as implemented in the pyDMS Python package (https://github.com/radosuav/pyDMS) (Figure 10). It is a quite complex method, which however works well even for sharpening by a ratio of 50 (i.e. from 1 km to 20 m). In the case of ETA product, the LST sharpening needs to be done by a ratio of around 3 (i.e. from 1 km to 300 m). This DMS was applied to sharpening of Sentinel-3 LST using Sentinel-2 reflectance by Guzinski and Nieto (2019), Guzinski et al. (2020) and Guzinski et al. (2023). Based on this work, we are applying it to sharpen Sentinel-3 LST product (**SL_2_LST___**) using Sentinel-3 Synergy product (**SY_2_SYN___**). SY_2_SYN___ product was used as a proxy for CLMS Sentinel-3 TOC reflectance product since the latter was still in production at the time this analysis was performed.

Briefly, the atmospherically corrected SYN product with a spatial resolution of 300 m is resampled to match the pixel sampling of the SLSTR thermal camera (around 1 km spatial resolution). Concurrently, Copernicus Digital Elevation Model (DEM) is used to derive slope and aspect maps which, together with Sentinel-3 overpass time, are used to estimate the solar irradiance incident angle of a tilted surface (i.e. accounting for terrain slope and exposure). The DEM and the solar angle maps are also resampled to the SLSTR resolution. A multivariate regression model is then trained with the three resampled datasets used as predictors and the SLSTR LST (T_{rad}) used as the dependent variable. Each low-resolution training sample is assigned a weight inversely proportional to the coefficient of variation (CV) of all the high-resolution pixels falling within one low-resolution pixel. The 20% most heterogeneous (i.e. with highest CV) samples are being extra penalized by having their weight halved. The regression model is based on bagging ensemble of decision trees. The decision trees are additionally modified such that all samples within a regression tree leaf node are fitted with a multivariate linear model, as proposed by Gao et al. (2012).

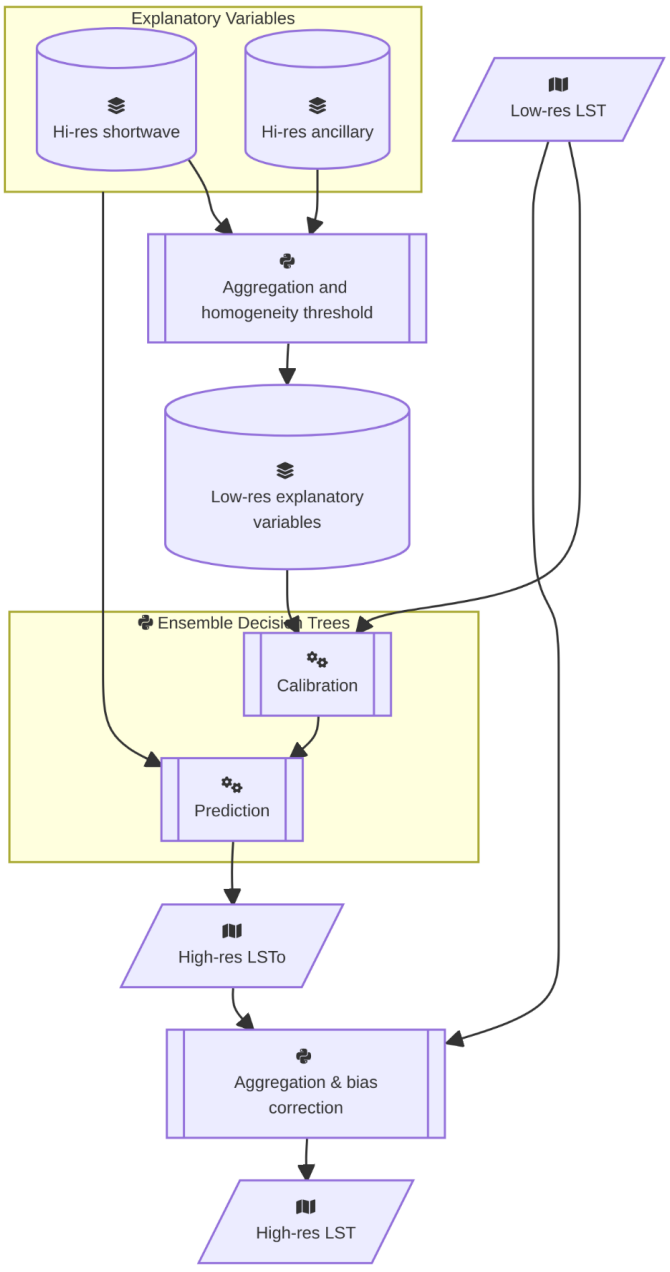


Figure 10: Data mining sharpener (DMS) workflow. Explanatory variables include multispectral bands as well as ancillary explanatory variable, such as elevation and solar angle maps. The regression model is based on an ensemble of modified decision trees.

The regression models are trained on the whole Sentinel-3 3-minute product data unit (PDU) (around 1400 km by 1200 km) as well as on subsets of 30 by 30 LST pixels in a moving window fashion. Once they are trained, they are also applied on the whole scene and on each window. The bias between the predicted high-resolution $\boldsymbol{T_{rad}}$ pixels aggregated to the low-resolution and the original low-resolution $\boldsymbol{T_{rad}}$ is calculated and the outputs of the whole-scene and moving-window regressions are combined based on a weight inversely proportional to the bias (Gao et al., 2012). Finally, the $\boldsymbol{T_{rad}}$ predicted by the regression model is corrected by comparing the emitted longwave radiance of the sharpened fine $\boldsymbol{T_{rad}}$ versus the original coarse $\boldsymbol{T_{rad}}$. A bias-corrected $\boldsymbol{T_{rad}}$ is therefore re-calculated by adding an offset all fine scale pixels falling within coarse scale pixel in order to remove any residual bias. This is done to ensure the conservation of energy between the two thermal images with different spatial resolutions (Gao et al., 2012). The output of the sharpening is a 300 m representation of the LST.

In the current framework, we use as explanatory variables: Oa04 (blue), Oa17 (NIR), Modified Normalized Difference Water Index (MNDWI, Xu, H., 2006), Plant Senescence Reflectance Index (PSRI, Merzlyak, et al. 1999), Normalized Difference Moisture Index (NDMI, Hardisky et al. 1983), Visible Atmospherically Resistant Index Red Edge (VARI_RED_EDGE, Ahamed et al. 2011), Bare Soil Index (BI, Nguyen et al. 2021), elevation, and cosine solar zenith angle.

3.2.7.5 Structural ancillary parameters

The effective transport of sensible heat can be computed as (Eq. 19):

$$H = \rho_{air} C_p \frac{T_0 - T_a}{r_a} \qquad \text{Eq. 19}$$

where ρ_{air} and C_p are the density of air and its specific heat respectively, and they can be assumed to be practically constant. T_0 (K) is the aerodynamic temperature, which *should not be confused with the radiometric surface temperature that is estimated with thermal infrared sensors*. T_a (K) is the air temperature measured at a height above ground z_T, and *significantly above the canopy height (h_c)*. r_a is the aerodynamic resistance to heat transport (s.m^{-1}), a reciprocal of which is the aerodynamic conductance.

This fundamental equation can be interpreted in a way that the heat exchange between the land surface and the atmosphere (H) is driven by a gradient of temperatures between the surface and the nearest layers of air above the canopy. This transport or exchange of heat is modulated by the

aerodynamic resistance, which includes a series of complex mechanical and convective processes that enhance or impede the transport of heat between the surface and the air.

Therefore, Resistance Energy Balance Models need additional ancillary inputs, such as canopy height or roughness. The latter accounts for the efficiency in the turbulent transport of heat and water between the land surface and the overlying air (Raupach, 1994; Alfieri et al., 2019). Vegetation structure and density are thus important for estimating turbulent transport of momentum, heat and water vapor in the canopy air space (Thom, 1972; Garratt and Hicks, 1973; Shaw and Pereira, 1982; Raupach, 1994).

Ideally, depending on the plant functional trait, roughness can be estimated using different methods. For short and herbaceous crops, the typical ratio of canopy height (h_c) can be applied (Eq. 20):

$$d_0 = 2/3\, h_c$$
$$z_{0m} = 1/8\, h_c \qquad \text{Eq. 20}$$

where d_0 is the zero-plane displacement height and z_{0m} is the roughness length for momentum.

For tall and woody canopies, in which leaf density and coverage can play a more important role, we could adopt the formulation of Schaudt (2000), which at the same time it relies on the model by Raupach (1994).

The CLMS global Dynamic Land cover map is used to assign ET model parameters, which are difficult to estimate directly from other Earth Observation data (Guzinski et al., 2021). Those parameters, and values assigned to different land cover classes, are listed in Table 9. Nevertheless, the TSEB-PT model requires all of the parameters, apart from stomatal resistance, while ETLook requires only vegetation height and stomatal resistance. Values of other parameters were adapted from Guzinski et al. (2020). In such framework, and for annual plant functional types, canopy height is computed considering its growth as (Eq. 21):

$$h_c = h_{min} + (h_{max} - h_{min}) min\left(\frac{LAI}{LAI_{max}}, 1\right) \qquad \text{Eq. 21}$$

However, in this case, in order to better adjust the obstacle (canopy) height in forests between the different climatic conditions, we use the ETH Zürich high-resolution canopy height model, 2020. A global 10m spatial resolution global forest canopy height map for 2020 that combines the Global Ecosystem Dynamics Investigation (GEDI) LiDAR and Sentinel-2 time-series (Lang et al., 2023).

Table 9: Land Cover (LC) Look-Up-Table for ancillary and structural parameters required by ET models, adapted from Guzinski et al. (2020), h_{min} (m) is the minimum canopy height; h_{max} (m) is the maximum canopy height occurring when LAI reaches its optimal maximum LAI_{max}; f_c is the at-Nadir fraction of the ground occupied by a clumped canopy (f_c=1 for a homogeneous canopy); w_c/h_c is the canopy shape parameter, representing the canopy width to canopy height ratio; l_w (m) is the average leaf size; α_{PT} is the initial Priestley-Taylor value for potential transpiration in TSEB; and r_{st} (s m^{-1}) is the minimum stomatal resistance. LC classes from 20 to 100 come from the CLMS global dynamic LC map while the LC classes from 111 to 126 come from the ETH Zürich Forest Canopy Height map.

LC	hc_{max} m	hc_{min} m	LAI_{max} m	f_c -	w_c/h_c -	l_w m	α_{PT}	R_{st} s.m^{-1}	description
20	2.0	2.0	N/A	1.0	1.0	0.05	1.26	175	Shrubs
30	0.1	1.0	4	1.0	1.0	0.02	1.26	175	Herbaceous vegetation
40	0.0	1.0	5	1.0	1.0	0.02	1.26	125	Cultivated and managed vegetation/agriculture (cropland)
50	10.0	10.0	0	0.0	0.0	0.0	1.26	400	Urban / built up
60	0.1	0.1	N/A	0.1	1.0	0.01	1.26	100	Bare / sparse vegetation
70	0.0	0.0	N/A	0.0	0.0	0.0	NA	100	Snow and ice
80	0.0	0.0	N/A	0.0	0.0	0.0	NA	100	Permanent water bodies
90	0.0	2.0	5	1.0	1.0	0.1	1.26	150	Herbaceous wetland
100	0.3	0.3	N/A	1.0	1.0	0.005	1.26	150	Moss and lichen
111	GEDI			0.8	0.5	0.05	Eq. 5	300	Closed forest, evergreen needle leaf
112	GEDI			0.8	1.0	0.15	0.82	180	Closed forest, evergreen, broad leaf
113	GEDI			0.8	0.5	0.05	Eq. 5	300	Closed forest, deciduous needle leaf
114	GEDI			0.8	1.0	0.15	0.82	190	Closed forest, deciduous broads leaf
115	GEDI			0.8	0.75	0.10	0.82	200	Closed forest, mixed
116	GEDI			0.8	0.75	0.10	0.82	180	Closed forest, unknown
121	GEDI			0.3	0.5	0.05	Eq. 5	300	Open forest, evergreen needle leaf
122	GEDI			0.3	1.0	0.15	0.82	180	Open forest, evergreen, broad leaf
123	GEDI			0.3	0.5	0.05	Eq. 5	300	Open forest, deciduous needle leaf
124	GEDI			0.3	1.0	0.15	0.82	190	Open forest, deciduous broads leaf
125	GEDI			0.3	0.75	0.10	0.82	200	Open forest, mixed
126	GEDI			0.3	0.75	0.10	0.82	180	Open forest, unknown
200	0.0	0.0	0.0	0.0	0.0	0.0	NA	0.0	Open sea

3.3 EVALUATION OF THE ALGORITHM PERFORMANCE

A reference database of Eddy Covariance (EC) sites was built in order to perform a direct validation against in situ measurements of ET. The priority was given to datasets were the Latent heat flux (LE) and the Sensible heat flux (H) had been corrected for the energy balance closure (EBC) problem. A commonly applied procedure delivering corrected values of LE and H is the ONEflux processing pipeline (https://github.com/fluxnet/ONEFlux). Data generated by this procedure could be obtained from the AmeriFlux (Novick et al 2018), ICOS (Heiskanen et al. 2022) and OzFlux (Isaac, 2014) networks. In search for more diversity in the reference datasets, other networks/datasets were queried even if the processing chain was not accounting for the EBC. Thus, data from EFDC (https://www.europe-fluxdata.eu/), AMMA (Galle et al. 2018), SAEON (https://www.saeon.ac.za/) and AsiaFlux (Mizoguchi, 2009) networks were also included in the analysis. Figure 11 shows the location of the eddy covariance sites considered in this study and the network/dataset the data were taken from.

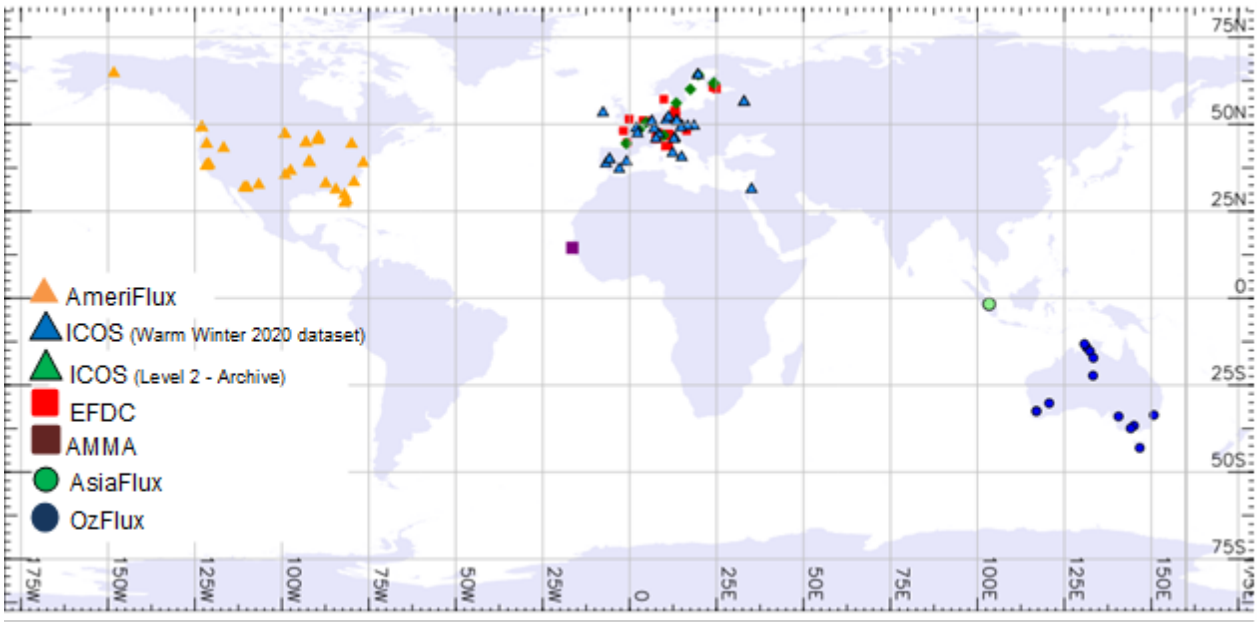


Figure 11: Eddy covariance sites and networks used for evaluating the models' performances

3.3.1 Dekadal ETA values

The models were evaluated over these sites for year 2020, which resulted in a dataset of 3108 records (104 sites) of dekadal ETa values obtained from eddy covariance measurements. Table 10 presents the overall validation results, for either the individual models (TSEB-PT and ETLook) and their ensemble (computed from Eq. 8). In addition, for comparison purposes, the dekadal ETa product from WaPOR Level 1 version 3 (https://data.apps.fao.org/catalog//iso/7f4a7339-d56e-4393-8712-a8ffeffe273) is also included in the validation.

Table 10: RMSE (mm/day), bias (mm/day) and r^2 scores for dekadal ETA per model, all sites together and summary values of statistical metrics at site level. rRMSE and rBias are the relative RMSE and relative Bias (i.e. RMSE and bias divided by mean measured ET).

	N	RMSE	Bias	rRMSE	rBias	r^2	Summary scores at site level					
							Bias		RMSE		r^2	
							min	max	min	max	mean	STD
TSEB-PT	3108	0.88	-0.12	0.52	-0.07	0.78	-1.19	1.41	0.28	1.77	0.76	0.25
ETLook	3108	1.08	-0.44	0.64	-0.26	0.78	-0.72	2.56	0.27	2.82	0.79	0.24
Ensemble	3108	0.80	-0.28	0.47	-0.16	0.84	-0.66	1.98	0.28	2.13	0.82	0.21
WaPOR	3108	0.85	0.12	0.50	0.07	0.81	-1.01	1.53	0.20	1.85	0.77	0.28

The points to be highlighted in these statistical scores are:

- The modelled ETa series under consideration exhibit high correlation with the reference dataset (r^2 around or above 0.8).
- The RMSE of the *Ensemble* is the lowest in the evaluation and the TSEB-PT yields the lowest absolute bias.
- The mean bias values of ETLook, TSEB-PT and the *Ensemble* ETa are negative

A complementary view on the model performance when grouping the sites per climate region is presented in the Taylor plots of Figure 12. Those plots show that the highest performance of all models is exhibited in the sites of the Temperate and Continental regions. It is important to note that the abundance of sites in these two climate regions is much higher than that of the Tropical

and Dry regions. The plots of Figure 12 point at the *Ensemble* and TSEB-PT dekadal ETa as the most suited options to better represent the different climate regions.

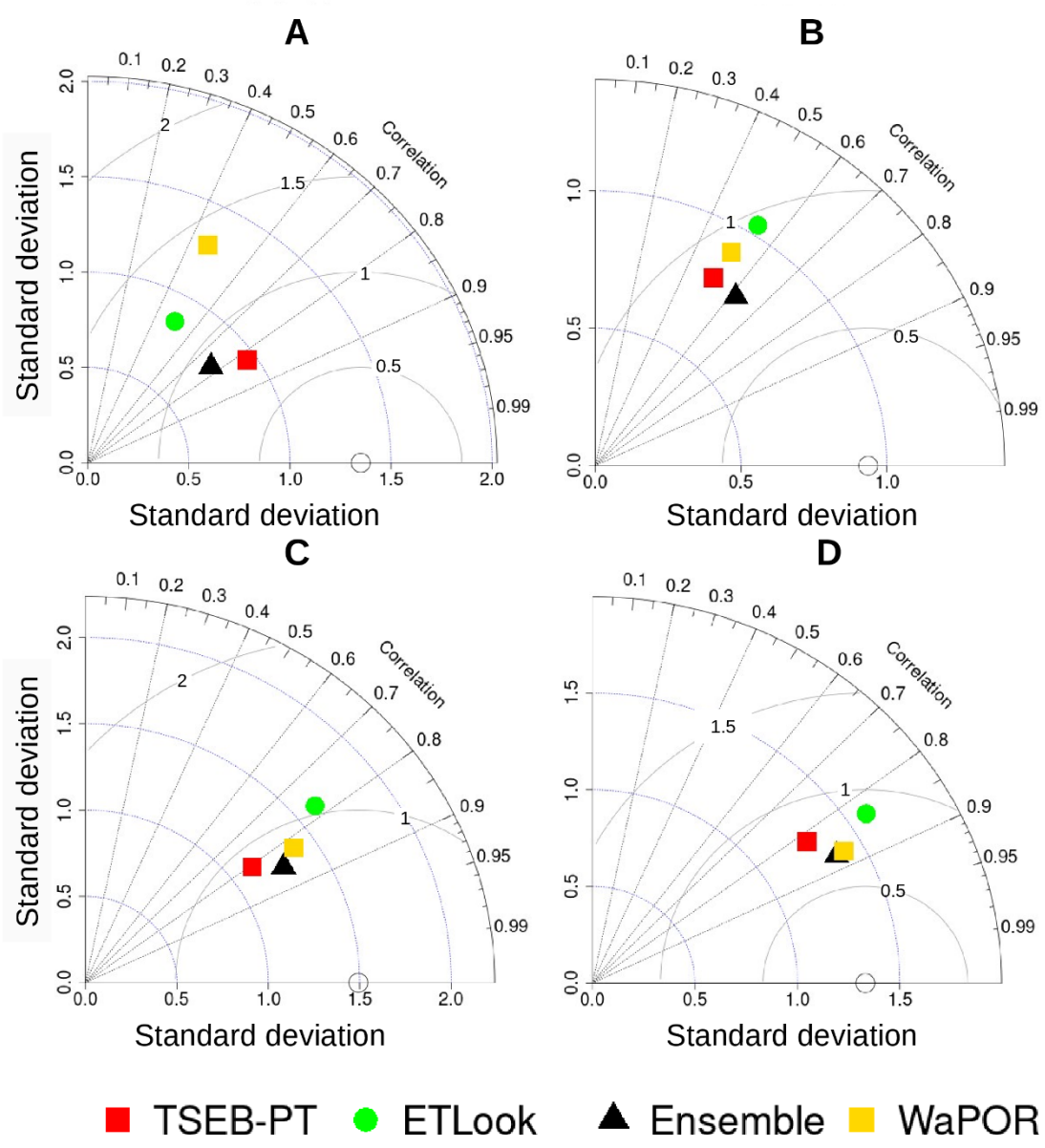


Figure 12: Taylor plots on the performance of the dekadal ET from TSEB-PT, ETLook, the *Ensemble* dataset and WaPOR in 2020. Sites grouped per Climate region. A: Tropical, B: Arid, C: Temperate, D: Continental.

Figure 13 also show that the performance in dekadal ETa modelling can often be associated to the plant functional type (PFT). Although the analysis per PFT can offer valuable insights on the performance of the models, one has to be cautious as one PFT class can contain a large diversity of ecosystem conditions.

The cropland (CRO), grassland (GRA) and wetlands (WET) plots of Figure 13 reveal that both models exhibited similar performance in the generation of dekadal ETA values in those groups. The three classes together account for 42% of the evaluated sites spread across different climate regimes and represent a wide range of conditions. The *Ensemble* ETA series exhibits slightly better correlation scores.

In savannas (SAV, WSA) and shrublands (CSH, OSH) biomes, more pronounced differences between the TSEB-PT and ETLook emerge. SAV sites (which are well represented in the dataset) show that TSEB-PT tends to underestimate the variability of the fluxes, whereas it is generally overestimated by ETLook. These biases seem to compensate each other in the *Ensemble* model, furthermore achieving a higher correlation than TSEB-PT and ETLook separately.

This behavior is even more pronounced in the forest sites (Deciduous Broadleaf Forest, Evergreen Broadleaf Forest, Evergreen Needleleaf Forest): TSEB-PT underestimates the variability of the flux (i.e. the amplitude of the seasonal cycle), whereas it is overestimated by ETLook.

The sites in the DBF group are all located in the Temperate or Continental climate region. The DBF group –with 10 sites- exhibit larger differences in the scores for TSEB-PT and ETLook. In the DBF class, the estimates of both models seemed to complement each other as ETLook estimates correlate better with the reference dataset than TSEB-PT but the error of the latter is smaller. In this case, the *Ensemble* model appears as a suited option for generating estimates closer to the reference ETA values.

The ETA in urban areas (URB) seems particularly difficult to reproduce by the models (which are not designed for this particular land cover type). The URB plot in Figure 13 indicates that the TSEB-PT and ETLook output correlate poorly with the reference data in urban areas and the difference between the two models is substantial.

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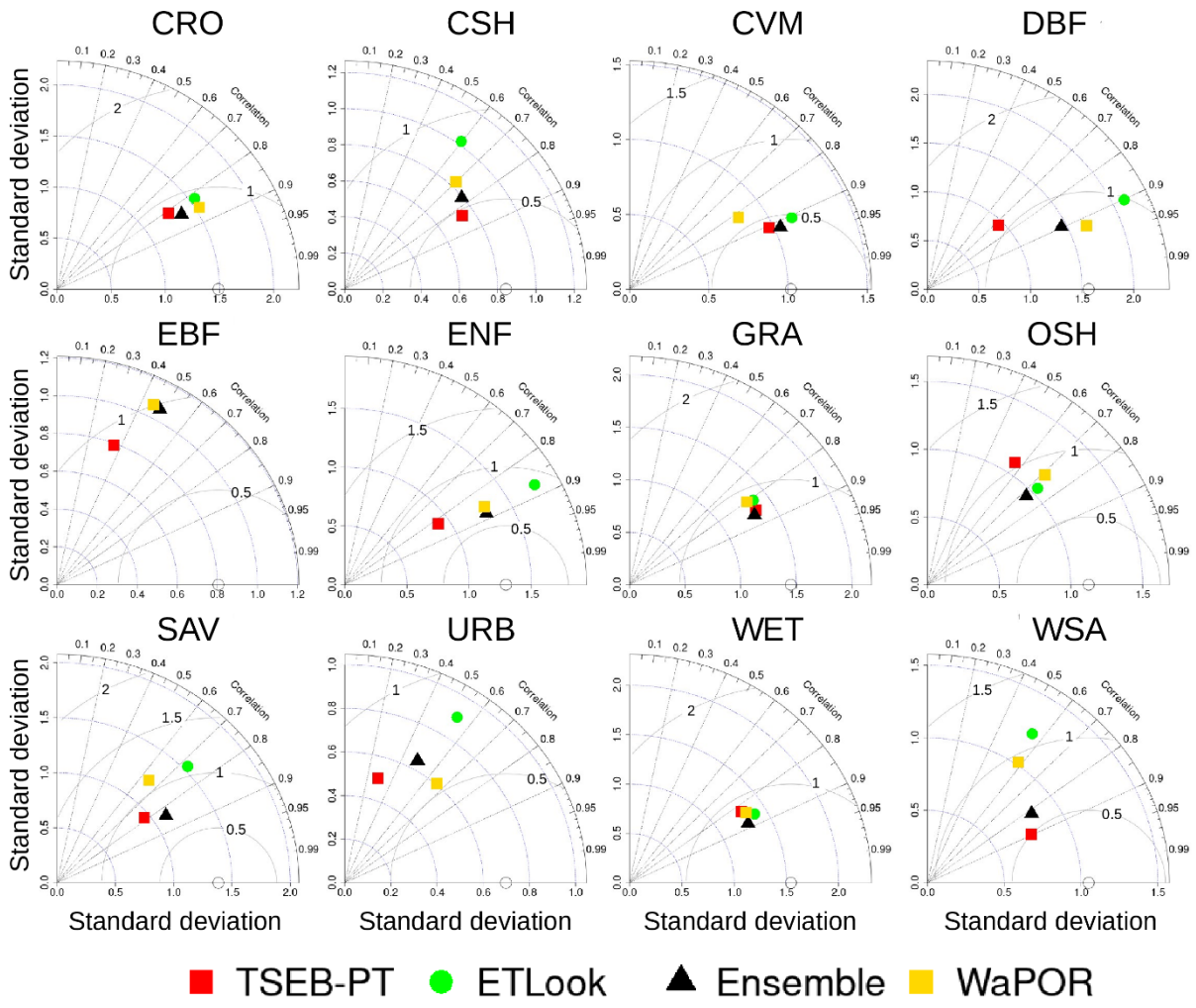


Figure 13: Taylor plots on the performance of the dekadal ET from TSEB-PT, ETLook, the *Ensemble* dataset and WaPOR in 2020. Sites grouped per PFT.

3.3.2 LE and H at satellite overpass time

Estimates of LE and H at satellite overpass time were generated by the TSEB-PT algorithm only. Therefore, this section is only intended to provide an overview of the goodness of fit of the modelled instantaneous LE and H estimates as compared with the values at the closest half-hourly timeslot reported at the eddy covariance towers. The satellite overpass time, in local time, varied along the year for the different sites.

The Figure 14 shows the range of r^2 and bias values obtained for each PFT class. A number of aspects in this plot can be connected to the analysis of dekadal ETA presented in the previous section. For instance, the low r^2 of LE in the urban sites and, to a certain degree, in the evergreen forest classes, the latter probably due to a lower ET dynamic range in those sites (typically semiarid).

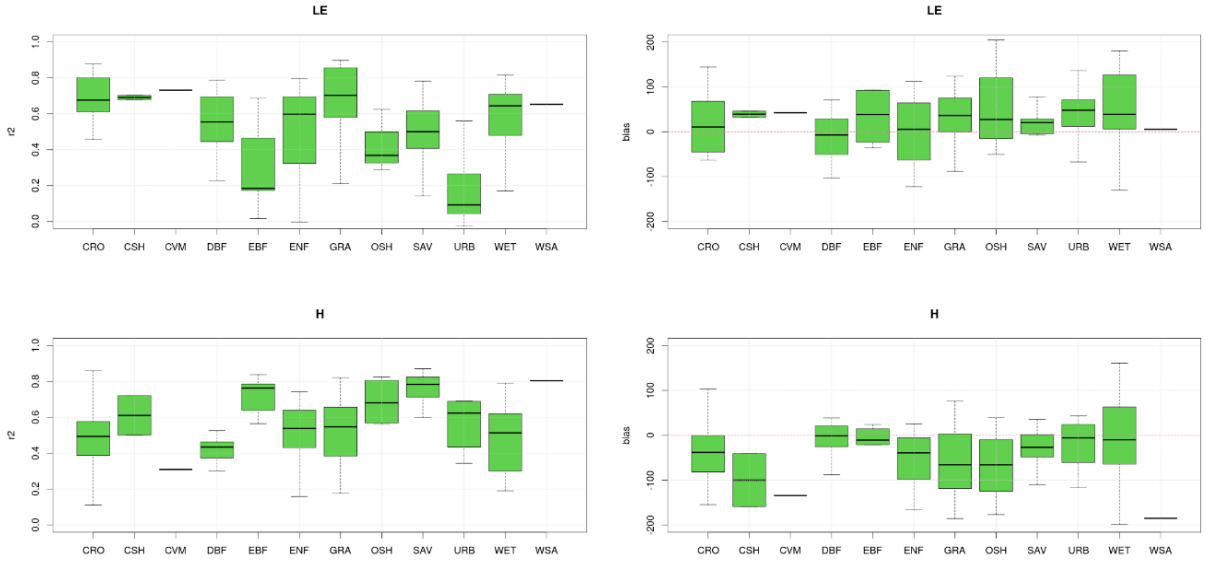


Figure 14: Boxplots of r^2 (left) and bias (right) values of LE and H at overpass time as modelled by the TSEB-PT model during the year 2020.

The length of the boxes in the boxplots of Figure 14 suggests an important degree of variability in the correlation and error of the modelled values amongst the sites of each PFT class. This is explained by the fact that some plant functional sites typically thrive under certain climatic conditions. That is the case of savannas, open shrublands, and evergreen broadleaved forests, which are typical of arid and semi-arid climatic zones (i.e. water-limited areas). Those cases are characterized by lower ETa rates, and thus lower LE fluxes that are closer to the actual model sensitivity/noise level, being more difficult to explain all the LE variability thus showing lower correlation values, while, on the opposite, the larger H fluxes in these areas tend to show much better correlations. On the other hand, PFTs that are typically on moist and energy-limited climates, such as DBF and WET, show a lower (higher) dynamic range of H (LE) values, with consequently

lower (higher) correlations. For this reason, it is worth noting that any validation of water and energy fluxes should include both LE and H together in the analysis.

3.4 LIMITATIONS OF THE ALGORITHM

The algorithm is limited to land pixels. Therefore, no evaporation data is provided for oceans and water bodies. Inland water pixels are identified by Bit 1 of the Flag (Table 1)

Furthermore, both ETLook and TSEB-PT models were not designed for urban surfaces, in which vegetated patches could co-exist with building and other built-up structures:

- Buildings induce additional turbulence and thus heat and water exchange between the land and the atmosphere is affected
- Also, they cast shadows over the lower vegetated/park areas and thus the radiative regime of vegetation is also affected
- There is a large heterogeneity in terms of both structural patterns and spectral features (i.e. albedo) at scales higher than 300m.

Therefore, we discourage its use in built-up land covers. Urban pixels are identified by Bit 2 of the Flag (Table 1).

3.5 RISK OF FAILURE AND MITIGATION MEASURES

Table 11: Identified risks and associated mitigation measures

Risk	Likelihood	Impact	Mitigation measure
Missing Sentinel-3 SLSTR LST	Low	High	Use LST from alternative platforms (VIIRS, METimage)
Missing CLMS TOC reflectance	Low	Moderate	Use of biophysical traits from previous days Use of global CLMS NRT LAI product Use data from alternative platforms (VIIRS, METimage)

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